CHAPTER

7

Quality Assurance and Validation

IMPORTANCE OF QUALITY ASSURANCE

Any scientific test that results in information that may lead to the loss of liberty of an individual accused of a crime needs to be performed with utmost care. DNA typing is no exception. It is a multi-step, technical process that needs to be performed by qualified and effectively trained personnel to ensure that accurate results are obtained and interpreted correctly. When the process is conducted properly, DNA testing is an effective investigative tool for the law enforcement community with results that stand up to legal scrutiny in court. When laboratories have not followed validated protocols or have not had adequately trained personnel, problems have arisen in the past.

Two topics are commonly referred to when discussing the importance of maintaining good laboratory practices to obtain accurate scientific results: quality assurance and quality control. *Quality assurance* (QA) refers to those planned or systematic actions necessary to provide adequate confidence that a product or service will satisfy given requirements for quality. *Quality control* (QC), on the other hand, usually refers to the day-to-day operational techniques and the activities used to fulfill requirements of quality.

Thus, an organization plans QA measures and performs QC activities in the laboratory. The forensic DNA community has long recognized the importance of quality results and, since early in the development of forensic DNA technology, has established organizations to recommend and oversee quality assurance guidelines and quality control measures.

Organizations Involved

A number of organizations exist around the world that work on a local, national, or international level to aid in quality assurance work and to ensure that DNA testing is performed

Advanced Topics in Forensic DNA Typing: Methodology.
DOI: 10.1016/B978-0-12-374513-2.00007-5

© 2012 Elsevier Inc. All rights reserved.

properly. The organizations are made up primarily of working scientists who want to coordinate their efforts to benefit the DNA typing community as a whole.

The *International Society of Forensic Genetics* (ISFG), which was founded in 1968 and formerly known as the International Society of Forensic Haemogenetics (ISFH), today represents a group of approximately 1100 scientists from more than 60 countries. Meetings are held biannually to discuss the latest topics in forensic genetics. Conference volumes were originally published under the title "Advances in Forensic Haemogenetics" and are now titled "Progress in Forensic Genetics." The 2007 ISFG meeting proceedings (*Progress in Forensic Genetics 12*) and 2009 ISFG meeting proceedings (*Progress in Forensic Genetics 13*) were published online as part of the *Forensic Science International: Genetics Supplement Series* (http://www.fsigeneticssup.com/).

Every few years, as a specific need arises, a *DNA Commission of the ISFG* is formed and makes recommendations on the use of genetic markers. These publications are available at http://www.isfg.org/Publications/DNA+Commission and include the following topics (with their publication year): DNA polymorphisms (1989), PCR-based polymorphisms (1992), naming variant STR alleles (1994), STR repeat nomenclature (1997), mitochondrial DNA (2000), Y-STR use in forensic analysis (2001), additional information on Y-STRs including nomenclature (2006), mixture interpretation (2006), disaster victim identification (2007), biostatistics for paternity testing (2008), and non-human (animal) DNA testing (2011). For additional information on ISFG, visit its website: http://www.isfg.org.

The *International Organization for Standardization* (ISO) issues guidance documents on a variety of topics. ISO 17025:2005 entitled "General requirements for the competence of testing and calibration laboratories" is the standard to which many DNA testing laboratories seek to be accredited by accrediting organizations, such as ASCLD/LAB or FQS-I (see below). For more information on ISO, see http://www.iso.org.

The *Technical Working Group on DNA Analysis Methods* (TWGDAM) was established in November 1988 under FBI Laboratory sponsorship to aid forensic DNA scientists throughout North America. Since 1998, TWGDAM has been known as SWGDAM, which stands for the *Scientific Working Group on DNA Analysis Methods*. SWGDAM is a group of approximately 50 scientists representing federal, state, and local forensic DNA laboratories in the United States and Canada. Meetings are held twice a year, usually in January and July. Public meetings have also been held in conjunction with scientific meetings such as the International Symposium on Human Identification, sponsored each fall by the Promega Corporation. Since 2006 a public SWGDAM meeting has been held as part of the FBI-sponsored National CODIS Conference.

Over the past two decades, five individuals have served as TWGDAM or SWGDAM chairmen: James Kearney (FBI), Bruce Budowle (FBI), Richard Guerrieri (FBI), David Coffman (Florida Department of Law Enforcement), and Ted Staples (Georgia Bureau of Investigation). In January 2011, Tony Onorato (FBI) was appointed by the FBI Laboratory Director to be the chair of SWGDAM.

Over the years, a number of TWGDAM or SWGDAM subcommittees have operated to bring recommendations before the entire group. These subcommittees have included (at different times): restriction fragment length polymorphism (RFLP), polymerase chain reaction (PCR), Combined DNA Index System (CODIS), mitochondrial DNA, short tandem repeat (STR) interpretation, training, validation, Y-chromosome, expert systems, quality assurance,

missing persons/mass disasters, mixture interpretation, mass spectrometry, enhanced method detection and interpretation, and rapid DNA analysis. TWGDAM issued guidelines for quality assurance in DNA analysis in 1989, 1991, and 1995. Revised SWGDAM validation guidelines were published in 2004 (these will be discussed in more detail later in the chapter). Several ad hoc subcommittees have produced recommendations on such topics as review of outsourced data and handling partial matches. The reference list contains all published SWGDAM guidelines as of early 2011.

The *DNA Advisory Board* (DAB) was a congressionally mandated organization that was created and funded by the United States Congress DNA Identification Act of 1994. The first meeting of the DAB was held on May 12, 1995 and chaired by Nobel laureate Dr. Joshua Lederberg. The DAB consisted of 13 voting members that included scientists from state, local, and private forensic laboratories; molecular geneticists and population geneticists not affiliated with a forensic laboratory; a representative from the National Institute of Standards and Technology (NIST); the chair of TWGDAM; and a judge.

The original members of the DAB included Joshua Lederberg (Rockefeller University), Arthur Eisenberg (University of North Texas Health Science Center), Shirley Abrahamson (Wisconsin State Supreme Court), Jack Ballantyne (Suffolk County Crime Lab), Bruce Budowle (FBI Laboratory), Ranajit Chakraborty (University of Texas Health Science Center), Bernard Devlin (Carnegie Mellon University), Marcia Eisenberg (Laboratory Corporation of America), Paul Ferrara (Virginia Division of Forensic Science), John Hicks (Alabama Department of Forensic Sciences), Margaret Kuo (Orange County Sheriff's Office), Terry Laber (Minnesota Bureau of Criminal Apprehension), and Dennis Reeder (NIST). A number of other individuals attended and participated in various DAB meetings over the five years that the group met.

The DAB was created for a five-year period to issue standards for the forensic DNA community. When the DAB's responsibilities ended in 2000, SWGDAM was designated as the group responsible for offering recommendations to the forensic community within the United States. In 2007 SWGDAM revised the Quality Assurance Standards for Forensic Caseworking Laboratories and the Quality Assurance Standards for DNA Databasing Laboratories. These revised standards went into effect July 1, 2009 after being approved by the FBI Laboratory Director. Further revisions regarding data review were made at the January 2011 SWGDAM meeting.

The *American Society of Crime Laboratory Directors* (ASCLD) and its Laboratory Accreditation Board (ASCLD/LAB) play an important role in the United States as well as internationally for laboratory accreditation programs. The ASCLD/LAB motto is "quality assurance through inspection." The Crime Laboratory Accreditation Program is a voluntary program in which any crime laboratory may participate to demonstrate that its management, operations, personnel, procedures, and instruments meet stringent standards. The goal of accreditation is to improve the overall service of forensic laboratories to the criminal justice system. If a forensic laboratory is interested in becoming accredited, an ASCLD/LAB Accreditation Manual is available from the Executive Secretary for a fee. Laboratories becoming accredited in forensic biology are audited against the FBI's Quality Assurance Standards for the laboratory operations pertaining to DNA testing. ASCLD/LAB accreditation may be under the Legacy Program or the International Program. However, no new Legacy Program applications have been processed since April 2009. Laboratories are being supported under

the Legacy Program until they can transition to ASCLD/LAB-*International*, which accredits to ISO/IEC 17025 requirements. As of December 2010, a total of 388 crime laboratories were accredited by ASCLD/LAB although not all of them are doing DNA testing. For additional information on ASCLD/LAB, visit its website: http://www.ascld-lab.org.

Forensic Quality Services (FQS) and *Forensic Quality Services – International* (FQS-I) are not-for-profit organizations established in 2003 by the National Forensic Science Technology Center (NFSTC), which is located in Largo, FL. FQS-I accredits forensic laboratories to ISO 17025. As of December 2010, a total of 52 labs were accredited by FQS-I. For additional information on FQS and FQS-I, visit their website: http://www.forquality.org.

The *American Association of Blood Banks* (AABB) sets standards for laboratories performing DNA parentage or relationship testing. AABB provides accreditation for paternity testing laboratories. As of December 2010, there were 39 accredited paternity testing laboratories in the United States. For more information on AABB, visit its website: http://www.aabb.org.

The *American Board of Criminalistics* (ABC) offers voluntary levels of certification for forensic scientists. Exams are given on a regular basis to test general understanding or specific knowledge in an area of forensic science such as molecular biology or drug analysis. Two levels of certification can be achieved: Diplomat (D-ABC) and Fellow (F-ABC). For more information on ABC, visit its website: http://www.criminalistics.com.

The *European Network of Forensic Science Institutes* (ENFSI) was started in 1995 to set standards for exchange of data between European member states and to be an accrediting body through conducting laboratory audits. Within the ENFSI, there is a DNA working group that meets twice a year to discuss forensic DNA protocols and research in much the same fashion as SWGDAM does within North America. For additional information on ENFSI, visit its website: http://www.enfsi.eu.

The European forensic DNA community has another organization similar to SWGDAM named *European DNA Profiling Group* (EDNAP), that first began meeting in October 1988. EDNAP is effectively a working group of the International Society of Forensic Genetics and consists of representatives from more than a dozen European nations. EDNAP has conducted a series of interlaboratory studies on various STR markers to investigate the reproducibility of multiple laboratories in testing the same samples. These studies have demonstrated that with the proper quality control measures excellent reproducibility can be achieved between forensic laboratories. EDNAP typically meets twice a year in conjunction with the ENFSI DNA working group meetings. For additional information on EDNAP, visit its website: http://www.isfg.org/EDNAP.

A *Forensic Science Regulator* works with organizations and practioners who provide forensic science services to the criminal justice system in England and Wales. Following concerns in data quality raised by the Omagh bombing trial (see D.N.A. Box 11.1), the office of a Forensic Science Regulator was established in the UK. Andrew Rennison was appointed in February 2008 to a three-year term, which was renewed in early 2011. The Regulator oversees efforts to maintain or improve quality in UK forensic laboratories. For further information, see http://www.homeoffice.gov.uk/police/forensic-science-regulator/.

The *College of American Pathologists* (CAP) offers external proficiency testing to forensic and paternity testing labs as well as clinical laboratories. For further information on CAP, visit their website: www.cap.org.

Cellmark Diagnostics, now *Orchid Cellmark*, a forensic DNA testing laboratory, provides a proficiency test program to help ensure ongoing laboratory quality. Their International Quality Assessment Scheme (IQAS) DNA Proficiency Test Program is designed for all laboratories conducting forensic DNA analysis. The proficiency tests consist of simulated forensic evidence case samples that are distributed four times a year. The Cellmark tests include questioned bloodstain and semen stain evidence along with known samples of blood. For more information on Orchid Cellmark, visit their website: www.orchidcellmark.com/forensicdna/iqasproficiency.html.

Collaborative Testing Services, Inc. (CTS) is an ASCLD/LAB proficiency test provider offering six different tests in its forensic biology program. For more information on CTS, visit their website: www.collaborativetesting.com.

The *Human Identity Trade Association* (HITA) is a non-profit organization that represents the interests of DNA companies and suppliers within the human identity market. HITA generally meets in conjunction with the International Symposium on Human Identification each fall. For additional information on HITA, visit the organization's website: www.humanidentity.org.

The *National Institute of Standards and Technology* (NIST) develops Standard Reference Materials (SRMs) that may be used by forensic laboratories to calibrate and verify their analytical procedures. Under the DAB standards, a laboratory is required to check its DNA procedures annually or whenever substantial changes are made to the protocol(s) against an appropriate and available NIST Standard Reference Material or a standard traceable to a NIST standard (QAS Standard 9.5.5). The various SRMs available from NIST are described below in the section on DNA standards. For additional information regarding NIST, visit its website: www.nist.gov.

The *Serological Research Institute* (SERI) is an ASCLD/LAB proficiency test provider with body fluid identification and mock case proficiencies offered to forensic laboratories. For more information on SERI, visit their website: www.serological.com.

The *Grupo Iberoamericano de Trabajo en Analisis de DNA* (GITAD) was organized in 1998 to serve the needs of forensic DNA laboratories and institutions in Latin America and the two countries of the Iberian Peninsula, Spain and Portugal. Much like SWGDAM and ENSFI, the primary objectives of GITAD include standardizing techniques, implementing a quality assurance/quality control system, and facilitating communication and training of laboratory personnel. For additional information regarding GITAD, visit their website: gitad.ugr.es/principal.htm.

The *Senior Managers of Australian and New Zealand Forensic Laboratories* (SMANZFL) was formed in 1986 to operate in a similar fashion as ASCLD does in the U.S. SMANZFL is supported by Specialist Advisory Groups, of which there are eight. The *Biology Specialist Advisory Group* (BSAG) oversees DNA efforts and is essentially equivalent to the U.S. SWGDAM group. For more information on SMANZFL, visit their website: www.nifs.com.au/SMANZFL/index.asp.

The *Asian Forensic Sciences Network* (AFSN) was formed in 2008. The group has a quality assurance and standards committee and workgroups for DNA, illicit drugs, toxicology, and trace evidence. As of early 2011, member institutes include representatives from Brunei Darussalam, Indonesia, Korea, Malaysia, Mongolia, People's Republic of China, Philippines, Singapore, Thailand, and Vietnam. For more information on AFSN, visit their website: www.asianforensic.net.

LEVELS OF QA/QC

Table 7.1 summarizes the overall quality assurance measures utilized within the United States, from the community level down to the individual case, that are in place to help provide confidence in the results produced by laboratories. Each of these levels will be covered in more depth in subsequent sections.

At the broadest level of the entire U.S. forensic DNA typing community, *quality assurance standards* have been established that require laboratories to carefully conduct their work. Guidelines and recommendations from organizations like SWGDAM and the ISFG DNA Commission play an important role in community-wide standardization. *Exploratory interlaboratory studies*, although not required, are also occasionally conducted to demonstrate that laboratories are obtaining comparable results.

At the individual laboratory level, *audits* are conducted regularly and *accreditation* granted when minimum standards are met. *Proficiency testing* and continuing education are required to help the individual analyst stay current in his or her work. Individual instruments and methods are validated and performance verified on a regular basis—often with traceability to a certified reference material. Each laboratory has *standard operating procedures* to describe the steps of the process to be followed to help ensure consistency over time and across cases within that laboratory.

At the sample batch level, positive and negative PCR amplification controls help demonstrate that sample reagents are working properly. Allelic ladders verify that all detectable STR alleles can be resolved from one another. Internal size standards are included with each sample processed for capillary electrophoresis STR allele separation and detection. Technical and administrative case review is performed to confirm the results obtained and conclusions reached. Finally, within the U.S. legal system, defendants have a right to counsel. Defense discovery requests and review of case information on evidence going to court provides an additional quality check on the DNA results (see Chapter 18).

TABLE 7.1 Quality Assurance/Quality Control (QA/QC) Measures in Place at Each Level Within the Forensic DNA Community.

Level	QA/QC Measure
Community	Quality assurance standards
Laboratory	Accreditation and audits
Analyst	Proficiency tests and continuing education
Method/instrument	Validation of performance *(along with use of traceable standard samples)*
Protocol	Standard operating procedure written and followed
Data sets	Positive and negative amplification controls
Individual sample	Internal size standard present in every sample
Interpretation of result	Case review by a second qualified analyst or supervisor
Court presentation of evidence	Defense attorneys and experts (discovery requests)

Quality Assurance Standards

Forensic DNA laboratories in the United States are mandated by Congress to follow strict quality assurance standards. In October 1998, the FBI Laboratory's DNA Advisory Board issued *Quality Assurance Standards* (QAS) that define how forensic laboratories are required to conduct business. These QAS were revised a decade later and went into effect July 1, 2009 (QAS 2009). Some minor revisions were introduced in 2011 regarding data review. U.S. forensic DNA laboratories are governed by the QAS and regularly audited for their compliance to these standards.

There are 17 topics covered in the revised (and original) QAS: (1) scope, (2) definitions, (3) quality assurance program, (4) laboratory organization and management, (5) personnel, (6) facilities, (7) evidence/sample control, (8) validation, (9) analytical procedures, (10) equipment calibration and maintenance, (11) reports/documentation, (12) review, (13) proficiency testing, (14) corrective action, (15) audits, (16) safety, and (17) outsourcing.

Audits

A *laboratory audit* evaluates the entire operation of a laboratory. It is a systematic examination that may be conducted by the laboratory management or by an independent organization according to pre-established guidelines. A laboratory must possess *standard operating procedures* (SOPs) and adhere to these protocols. Likewise, instruments and other equipment vital to the successful completion of a forensic DNA case must be maintained properly, and personnel must be appropriately trained to perform their jobs.

An audit team, typically consisting of scientists from other forensic DNA laboratories, visits the lab being audited and evaluates this lab according to criteria spelled out in an audit/assessment document. Audit documents for forensic DNA testing and DNA databasing laboratories are available on the FBI website at http://www.fbi.gov/about-us/lab/codis/audit_testlabs and http://www.fbi.gov/about-us/lab/codis/audit-document-for-dna-databasing-labs, respectively.

Records of an audit are maintained and serve to describe the findings of the audit team and a course of action that may be taken to resolve any existing problems. If any negative findings are noted, then the lab has to take corrective action or lose accreditation.

Laboratory Accreditation

Laboratory accreditation results from a successful completion of an inspection or audit by an accrediting body. The two primary accrediting organizations within the United States that are recognized by the forensic DNA community are ASCLD-LAB and FQS (and FQS-I). Accreditation requires that the laboratory demonstrate and maintain good lab practices including chain-of-custody and evidence-handling procedures.

The accreditation process generally involves several steps such as a laboratory self-evaluation, filing application and supporting documents to initiate the accreditation process, on-site inspection by a team of trained auditors, an inspection report, and an annual accreditation review report. The inspection evaluates the facilities and equipment, the training of the technical staff, the written operating and technical procedures, and the casework reports and supporting documentation of the applicant laboratory.

Proficiency Tests

A *proficiency test*, as it relates to the DNA typing field, is an evaluation of a laboratory analyst's performance in conducting DNA analysis procedures. These tests are performed periodically, usually on a semiannual basis, for each DNA analyst or examiner. In fact, the Quality Assurance Standards (Standard 13.1) require that each DNA analyst undergo an external proficiency test at least twice a year. Biological specimens with a previously determined DNA profile are submitted to the laboratory personnel being tested. The purpose of the test is to evaluate the analyst's ability to obtain a concordant result using the laboratory's approved SOPs.

These proficiency tests may be administered by someone else in the laboratory (*internal proficiency test*) or by an external organization (*external proficiency test*). If the test administered by an external organization is performed such that the laboratory personnel do not know that a test is being conducted, then it is termed a *blind external proficiency test*. A blind external proficiency test is generally considered the most effective at monitoring a laboratory's abilities but can be rather expensive and time-consuming to arrange and conduct.

Participation in a proficiency-testing program is an essential part of a successful laboratory's quality assurance effort. Forensic laboratories develop their own proficiency-testing programs or establish one in cooperation with other laboratories. The German DNA profiling group (GEDNAP) has established a successful blind proficiency-testing program (Rand et al. 2002, Rand et al. 2004).

The purpose of proficiency testing is to evaluate the performance of an analyst using a sample or set of samples that is unknown to the analyst but known to the test provider. Recommendation 3.2 of 1996 National Research Council report *The Evaluation of Forensic DNA Evidence* states that: "Laboratories should participate regularly in proficiency tests, and the results should be available for court proceedings." Successful completion of this examination permits a degree of confidence to exist in how an analyst might perform on a real forensic case sample. Unfortunately, if analysts are aware that they are being tested, they might be more careful than they would when normally processing routine samples on a daily basis. Thus, the concept of blind proficiency has often been discussed in order to have a true test of the entire system because the analyst would not know that he or she was being tested. However, a number of challenges and costs are associated with blind proficiency tests (Peterson et al. 2003a, 2003b).

Validation

While work quality of analysts can be confirmed through proficiency tests, instruments used and methods performed are verified through validation. *Validation* refers to the process of demonstrating that a laboratory procedure is robust, reliable, and reproducible in the hands of the personnel performing the test in that laboratory.

A *robust method* is one in which successful results are obtained a high percentage of the time and few, if any, samples need to be repeated. A *reliable method* refers to one in which the obtained results are accurate and correctly reflect the sample being tested. A *reproducible method* means that the same or very similar results are obtained each time a sample is tested. All three qualities are important for methods performed in forensic laboratories.

There are generally considered to be two stages of validation: developmental validation and internal validation. *Developmental validation* involves the testing of new STR loci or STR

kits, new primer sets, and new technologies for detecting STR alleles. *Internal validation*, on the other hand, involves verifying that established procedures, previously developmentally validated will work effectively in one's own laboratory. Developmental validation is typically performed by commercial STR kit manufacturers and large laboratories such as the FBI Laboratory while internal validation is the primary form of validation performed in smaller local and state forensic DNA laboratories. Further details on developmental and internal validation will be covered later in this chapter.

Standard Operating Procedures and Interpretation Guidelines

Each forensic laboratory develops or adopts standard operating procedures (SOPs) that give a detailed listing of all the materials required to perform an assay as well as the exact steps required to successfully complete the experiment. In addition, SOPs list critical aspects of the assay that must be monitored carefully. SOPs are followed exactly when performing forensic DNA casework.

SOPs also include interpretation guidelines to aid analysts in making decisions in evaluating data. For example, should a peak observed in an electropherogram be called an allele or be removed from consideration in the DNA profile because it is an instrumental artifact (e.g., a spike) or a biological artifact (e.g., a stutter product)? These interpretation guidelines are based on internal validation studies performed by the laboratory. Usually, the more specific the interpretation guidelines are written, the more consistent data review will be between analysts within a laboratory and across cases examined by the same analyst over time.

Case Review

As with patients who request a second opionion when faced with a medical procedure, DNA results are reviewed by a second analyst and/or supervisor to confirm allele calls and case conclusions. Forensic DNA laboratories conduct administrative and technical reviews of case files and reports prior to releasing this information to ensure that conclusions reached are reasonable based on the data collected.

Controls for Data Sets and Individual Samples

Multiple levels of quality control are available for DNA samples being processed. Biological samples are quantified to determine the amount of human DNA present prior to PCR amplification. Reagent blanks, which are analytical control samples containing no template DNA, are carried through the DNA typing process from extraction onward to monitor potential contamination. PCR amplification positive and negative controls are amplified alongside the casework evidence or reference samples to demonstrate that the PCR reagents are working properly and are not contaminated (see Chapter 4).

Allelic ladder samples permit an evaluation of DNA separation, resolution, and sizing precision over time—particularly because multiple allelic ladders are often associated with each set of data collected. Each sample is also mixed with an internal size standard prior to being analyzed with capillary electrophoresis. These controls enable troubleshooting of the process should something go wrong during PCR amplification or data collection on the CE

instrument (see Chapter 6). Proper performance of each control also provides confidence in the results obtained.

Court Proceedings

A case report prepared based on the laboratory results obtained may be entered into evidence as part of court proceedings. In the United States' legal system, defense attorneys and defense experts have the power to make *discovery requests* (see Chapter 18). Under a discovery request, the forensic laboratory is required to turn over the items requested so that they can be reviewed in the interest of the defendant. Thus, lab results may be scrutinized before and during a trial. This provides another level of review to ensure that quality results have been obtained in the case under consideration. In addition, an analyst may be called upon to testify in court to the DNA results obtained. Cross-examination by the defense team then provides the final level of review in order to confirm the DNA testing results.

LEVELS OF VALIDATION

Validation is an important part of forensic DNA typing and provides confidence in results obtained. Judges and defense lawyers today rarely challenge the science behind DNA typing—rather they challenge the process by which the laboratory and the analyst performs the DNA analysis. Thus, the scientific community must carefully document the validity of new techniques and technologies to ensure that procedures performed in the laboratory accurately reflect the examined samples. In addition, a laboratory must carefully document their technical procedures and policies for interpretation of data and follow them to guarantee that each sample is handled and processed appropriately.

ISO 17025 Section 5.4.5.1 states: "Validation is the confirmation by examination and the provision of objective evidence that the particular requirements for a specific intended use are fulfilled" (ISO/IEC 17025:2005). Furthermore, the FBI QAS define validation as "a process by which a procedure is evaluated to determine its efficacy and reliability for forensic casework analysis" (QAS 2009). Thus, a primary purpose of validation is to demonstrate that a method is suitable for its intended purpose.

With the adoption of ISO 17025 by many forensic laboratories, analysts have to face the concept of *measurement uncertainty* (D.N.A. Box 7.1). One of the purposes of validation is to gain a better understanding of the measurement uncertainty associated with specific steps in the analytical process used to perform forensic DNA testing.

General levels of validation include the following:

- **Developmental Validation**—commonly performed by the commercial manufacturer of a novel method or technology (more extensive than internal validation)
- **Internal Validation**—performed by an individual lab when a new method is introduced
- **Performance Checks**—performed after instrument repair or receipt of new reagents to verify that the instrument or reagents are working properly; monitoring and assessment of control samples and the internal size standard run with each test or set of samples can be considered a performance check.

D.N.A. BOX 7.1

MEASUREMENT UNCERTAINTY

Limitations exist in our ability to measure anything. We can never be 100% certain in science. As Jeff Salyards notes in his article *Estimating the Uncertainty*: "Remembering the limitations of our measurements will keep us intellectually honest, and distinguishing opinions and interpretations will clarify our reports for the investigative and legal communities" (Salyards 2008). An important purpose of validation experiments is to better understand the limitations of measurements (e.g., sensitivity and reproducibility) being made in the laboratory with a specific method.

International Standards Organization (ISO) 17025 document Section 5.4.6 discusses "measurement uncertainty." This term can be confused with the term "error." While in measurement science the term "error" is the difference between the measured value and the true value of the thing being measured, "uncertainty" can best be summed up as a quantification of the doubt about the measurement result (Bell 1999). Uncertainty addresses the question of "Just how sure are you?" about a result produced when measuring something (Salyards 2008).

To measure uncertainty, the sources of uncertainty need to be known and quantified. Replicate analyses are typically performed and these measurements then statistically evaluated (Wallace 2010). The results may then be summarized with an average value plus or minus the standard deviation. Just including the average value does not reflect the level of variation being observed in a data set, which is why a standard deviation is appropriately included when listing an average value. Detailed guidance on measurement uncertainty has been published (EURACHEM 2000, GUM 2008).

A recent article summarizes ten methods for estimating this measurement uncertainty: (1) proficiency tests; (2) readability limits; (3) independent reference materials; (4) operational limits applied during calibration; (5) expert judgment; (6) precision control samples without (6) and with (7) contributions from extramural sources of error; (8) error budgets; (9) historical performance; and (10) ruggedness tests (Wallace 2010).

Measurement uncertainty exists with DNA testing particularly during the DNA quantitation and STR allele sizing steps. With appropriate validation studies, variation that does exist can be understood so that experiments performed may yield reliable results.

Sources:

EURACHEM/CITAC. (2000). Quantifying Uncertainty in Analytical Measurement. Available at http://www.citac.cc/QUAM2000-1.pdf.

ISO/IEC 17025:2005 (2005). General requirements for the competence of testing and calibration laboratories. Available from http://www.iso.org.

Joint Committee for Guides in Metrology (2008). Evaluation of Measurement Data – Guide to the Expression of Uncertainty in Measurement *(GUM). Available at http://www.bipm.org/en/publications/guides/gum.html.*

NIST Reference on Constants, Units, and Uncertainty: http://www.physics.nist.gov/cuu/Uncertainty/index.html.

Salyards, J. (2008). Estimating the uncertainty. Forensic Magazine (Feb/Mar 2008), *42–44. Available at http://www.forensicmag.com/article/estimating-uncertainty.*

Taylor, B.N., & Kuyatt, C.E. (1994). Guidelines for evaluating and expressing the uncertainty of NIST measurement results. NIST Technical Note 1297. *Available at http://physics.nist.gov/Pubs/guidelines/TN1297/tn1297s.pdf.*

Wallace, J. (2010). Ten methods for calculating the uncertainty of measurement. Science & Justice, 50, *182–186.*

History of Guidance Documents on Forensic DNA Validation

The primary purpose in forming TWGDAM and maintaining SWGDAM over the past two decades is to provide guidance on validation and other aspects of quality measurements and data interpretation in forensic DNA typing laboratories. As mentioned previously, TWGDAM issued guidelines for quality assurance in DNA analysis in 1989, 1991, and 1995. Revised SWGDAM validation guidelines were published in 2004. The original DAB quality assurance standards released in 1998 and 1999 have been superceded by the FBI revised QAS of 2009. (The validation section did not change for the revised 2011 QAS.) The ENFSI DNA Working Group issued guidance on validation in November 2010 (ENFSI 2010).

The topics covered in the 1991/1995 TWGDAM, 2004 SWGDAM, 1998/1999 QAS, and 2009/2011 QAS are compared for developmental validation studies (Table 7.2) and internal validation studies (Table 7.3). The 1989 TWGDAM guidelines are not included in these comparisons as the early guidelines focused on the now obsolete restriction fragment length polymorphism methodologies. Evolving technology and laboratory practices make it necessary to periodically issue revisions in quality assurance standards for DNA testing. While only the current guidelines or standards may be applicable to laboratories today, it is helpful to keep a historical perspective.

Developmental Validation Studies

Standard 8.2 in the 2009/2011 revised FBI Quality Assurance Standards requires that developmental validation precede the use of a novel methodology for forensic DNA analysis (QAS 2009). Over the years, different topics have been emphasized as part of developmental validation studies in different guidance documents (Table 7.2).

QAS Standard 8.2.1 incorporates the studies listed in the SWGDAM Revised Validation Guidelines (SWGDAM 2004). These "include, where applicable, characterization of the genetic marker, species specificity, sensitivity studies, stability studies, reproducibility, case-type samples, population studies, mixture studies, precision and accuracy studies, and PCR-based studies" (QAS 2009). PCR-based studies are defined to "include reaction conditions, assessment of differential and preferential amplification, effects of multiplexing, assessment of appropriate controls, and product detection studies." Each of these topics is explored further in the next section.

Validation Studies and Their Purposes

While the 1995 TWGDAM guidelines were the basis for most early validation studies on STR typing kits, more recent validation studies use information from a set of revised validation guidelines that were approved by SWGDAM in July 2003 and finally published and made publicly available in July 2004 (SWGDAM 2004). The 2004 SWGDAM Revised Validation Guidelines have been incorporated into the revised QAS (QAS 2009). The standard developmental validation studies that are conducted to become "SWGDAM Validated" are listed below with either the original 1995 TWGDAM Validation Guideline (Budowle et al. 1995) or 2004 SWGDAM Revised Validation Guideline (SWGDAM 2004) numerical headings shown in parentheses. The purpose of each study is also enumerated. As noted in the

TABLE 7.2 Comparison of Developmental Validation Topics in Current and Previous TWGDAM/SWGDAM Validation Guidelines and DNA Advisory Board (DAB)/FBI Quality Assurance Standards (QAS).

TWGDAM (1991/1995)	**DAB QAS (1998/1999)**	**SWGDAM (2004)**	**FBI QAS (2009/2011)**
Guidelines 4.1 to 4.4	*Standards 8.1.1 &8.1.2*	*Guidelines 2.1 to 2.10*	*Standards 8.2*
Locus characteristics determined & documented: inheritance, chromosomal location, detection, polymorphism (4.2)	Documentation exists & available which defines & characterizes the locus (8.1.2.1)	Genetic marker characterization (2.1): inheritance (2.1.1), chromosomal location (2.1.2), detection (2.1.3), polymorphism (2.1.4)	Genetic marker characterization
Standard specimens (4.1.5.1)			
Different tissues, same type (4.1.5.1)	Precision & accuracy (8.1.2)	Precision & accuracy (2.9)	Precision & accuracy studies
Consistency within & between labs (4.1.5.2)			
Minimum sample (4.1.5.10)	Sensitivity studies (8.1.2.2)	Sensitivity studies (2.3)	Sensitivity studies
Population studies (4.1.5.3)	Population distribution data (8.1.2.3)	Population studies (2.7)	Population studies
Reproducibility (4.1.5.4)	Reproducibility (8.1.2)	Reproducibility (2.5)	Reproducibility
Environmental studies (4.1.5.6)	Stability studies (8.1.2.2)	Stability studies (2.4)	Stability studies
Matrix studies (4.1.5.7)			
Non-probative evidence (4.1.5.8)	Non-probative evidence (8.1.3.1)	Case-type samples (2.6)	Case-type samples
Non-human studies (4.1.5.9)	Species specificity (8.1.2.2)	Species specificity (2.2)	Species specificity
Mixed specimen studies (4.1.5.5)	Mixture studies (8.1.2.2)	Mixture studies (2.8)	Mixture studies
On-site evaluation (4.1.5.11)			
PCR-based procedures (4.4)		PCR-based procedures (2.10)	PCR-based studies
Reaction conditions: thermocycling parameters & critical reagent concentrations (primers, polymerase and salts) needed for required specificity (4.4.1.3)		Reaction conditions: thermocycling parameters, primer concentrations, $MgCl_2$, DNA polymerase, & other critical reagents (2.10.1)	Reaction conditions
Number(s) of cycles to produce reliable results (4.4.1.4)			
Assessment of differential amplification (4.4.1.5)		Amplification assessment: differential, preferential, stochastic (2.10.2)	Amplification assessment: differential & preferential

(*Continued*)

TABLE 7.2 Comparison of Developmental Validation Topics in Current and Previous TWGDAM/SWGDAM Validation Guidelines and DNA Advisory Board (DAB)/FBI Quality Assurance Standards (QAS). (*Continued*)

TWGDAM (1991/1995)	DAB QAS (1998/1999)	SWGDAM (2004)	FBI QAS (2009/2011)
Multiplex amplification effects documented (4.4.1.6)		Multiplex effects & artifacts (2.10.3)	Multiplexing effects
		Positive & negative controls validated (2.10.4)	Appropriate controls
Establish appropriate standards for assessing alleles (e.g., size markers) (4.4.2.1)		Measurement standards for allele (2.10.5.1.1) or sequence data (2.10.5.1.2) characterization	Product detection studies
Characterization with hybridization (4.4.2.2)		Detection by hybridization (2.10.5.2)	
PCR primers of known sequence (4.1.1.1)		Publication of primer sequences not required (2.10)	
Validation results shared & available (4.1.5.12)	Developmental validation appropriately documented (8.1.1)	Peer-reviewed publication encouraged (1.2.1.2)	Required peer-reviewed publication of underlying principles (8.2.2)

SWGDAM 2004 Revised Validation Guidelines, some studies may not be necessary for a particular method.

Developmental validation is performed and often published by the manufacturer of a new DNA test, instrument, or software (see Table 7.5). If a forensic laboratory decides to use a DNA test, instrument, or software in a way not previously explored by the manufacturer, then a full scale developmental validation may be in order. For example, if an STR typing kit is designed and tested by the manufacturer to work down to approximately 125 pg with 28 cycles, then increasing the cycle number to 31 cycles to improve the kit sensitivity would warrant additional validation studies (Caragine et al. 2009) in order to characterize amplification reliability and the stochastic effects likely to arise with low-level DNA analysis (see Chapter 11).

Major points regarding developmental validation in the *2004 SWGDAM guidelines* or *1995 TWGDAM guidelines* are emphasized below with their corresponding section:

- ***Characterization of genetic markers*** (2.1) or *locus characteristics* (4.2). It is important to understand what is being measured and how a specific genetic marker is expected to perform when tested. The inheritance, chromosomal location, method of detection, and polymorphism should be known and documented. Examination of two- or three-generation family samples is commonly performed to measure genetic marker inheritance. The Human Genome Project has aided determination of genetic marker locations throughout the human genome. Genome information on other species is often available as well due to extensive genomic studies in recent years. Numerous population studies

TABLE 7.3 Comparison of Internal Validation Topics in Current and Previous TWGDAM/SWGDAM Validation Guidelines and DNA Advisory Board (DAB)/FBI Quality Assurance Standards (QAS).

TWGDAM (1991/1995)	DAB/QAS (1998)	SWGDAM (2004)	QAS (2009/2011)
Guideline 4.5	*Standard 8.1.3*	*Guidelines 3.1 to 3.7*	*Standards 8.3 to 8.5*
Known samples (4.5.1)	Known and non-probative evidence samples (8.1.3.1)	Known and non-probative evidence samples (3.1)	Known and non-probative evidence or mock evidence samples (8.3.1)
	Monitor and document reproducibility and precision (8.1.3.1)	Reproducibility and precision (3.2)	Reproducibility and precision (8.3.1)
Precision to establish match criteria (4.5.3)	Establish & document match criteria (8.1.3.2)	Match criteria (3.3)	
		Sensitivity and stochastic studies (3.4)	Sensitivity and stochastic studies (8.3.1)
		Mixture studies (3.5)	Mixture studies (8.3.1)
Contamination assessment (4.5.4)		Contamination & controls (3.6)	Contamination assessment (8.3.1)
Proficieny test samples (4.5.5)	Qualifying test (8.1.3.3)	Qualifying test (3.7)	Competency test (8.4)
			Documented and summarized (8.3.1)
			Approved by technical leader (8.3.1)
			Summary available at each site in a multi-laboratory system (8.3.1.1)
			Studies to define QA parameters and interpretation guidelines (8.3.2)
			Change in detection platform or test kit shall require internal validation studies (8.3.3)
New procedures must be compared to original procedures with identical samples when modified (4.5.2)			Modified procedure compared to original procedure using similar DNA samples (8.5)

enable characterization of the alleles and polymorphism present at the genetic markers of interest. Information resources such as the NIST STRBase website (http://www.cstl.nist.gov/biotech/strbase) have documented locus characteristics for commonly used STR loci (see Chapter 5).

- ***Species specificity*** (2.2) or *non-human studies* (4.1.5.9). The DNA typing system being evaluated is subjected to non-human DNA to see if other biological sources could interfere with the ability to obtain reliable results on samples recovered from crime scenes. Primates, such as gorillas and chimpanzees, are typically tested along with domestic animals, such as horses, cattle, dogs, and cats. Bacteria and yeast, which can be prevalent in sexual assault evidence, are also tested. Most STR loci used for human identity testing are primate-specific, that is, they amplify in gorillas and chimps but not dogs or cats. Bacteria, yeast, and most non-primates typically do not yield any detectable products with the STR kits currently available. The sex-typing marker amelogenin does amplify in a number of other species but with DNA fragments that are slightly smaller in size than the standard 106 and 112 bp for human X and Y alleles (Buel et al. 1995).
- ***Sensitivity studies*** (2.3) or *minimum sample* (4.1.5.10). The minimum quantity of genomic DNA needed to obtain a reliable result is typically determined by examining a dilution series of a sample with a known genotype. For example, 10 ng, 5 ng, 2 ng, 1 ng, 0.5 ng, 0.25 ng and 0.1 ng might be evaluated. Most protocols call for using at least 0.25 ng to 0.5 ng genomic DNA for PCR amplification to avoid allele dropout from stochastic effects during the PCR step or poor sensitivity during the detection phase of the analysis. Laboratories wanting to perform low-level DNA work should perform additional validation studies to assess the impact of stochastic effects (see Chapter 11).
- ***Stability studies*** (2.4). If environmental insults, chemical insults, or substrates could potentially impact results, then they should be evaluated using known samples to determine the effects of these factors. Past studies have shown that a correct full or partial profile may be obtained in most instances and thus these studies are not performed as commonly today as in the early days of DNA testing.
- ***Environmental studies*** (4.1.5.6). Samples of known genotype are environmentally stressed and examined to verify that the correct genotype is obtained. The environmental studies reflect the situations typical of a forensic case (i.e., exposure to sunlight, humidity, and temperature fluctuations).
- ***Matrix studies*** (4.1.5.7). Samples of known genotype are examined after contact with a variety of substrates commonly encountered in forensic cases. For example, blood and semen may be deposited on leather, denim, glass, metal, wool, or cotton as well as mixed with dyes and soil. DNA profiles from samples exposed to these substrates are carefully examined for non-specific artifacts and amplification failure at any of the loci studied.
- ***Reproducibility*** (2.5) or *consistency* (4.1.5.2). The measurement technique is evaluated repeatedly to assess the reproducibility of the method within and sometimes between laboratories. The power of DNA testing is only fully realized when results can be compared between laboratories in different areas or when offender samples present in DNA databases can be accurately matched with crime scene samples originating from that offender. Thus, results must be comparable across both distance and time. The use of internal sizing standards and allelic ladders has greatly improved the consistency of STR typing across laboratories.

- ***Case-type samples*** (2.6) or *non-probative evidence* (4.1.5.8). DNA profiles are obtained from existing samples that are part of forensic cases that have already been closed. These samples demonstrate that the DNA typing system being examined can handle real casework situations. In some situations, mock evidentiary samples may be created to simulate PCR inhibitors, DNA degradation, or other challenges commonly found in forensic specimens.
- ***Population studies*** (2.7). A set of anonymous samples that have been grouped by ethnicity is analyzed to determine allele frequencies for each major population group that exists in a forensic laboratory's vicinity. These allele frequencies are then used in reporting population statistics and calculating the probability of a random match.
- ***Mixture studies*** (2.8) or *mixed specimen studies* (4.1.5.5). The ability of the DNA typing system to detect the various components of mixed specimens is investigated. Evidence samples in forensic cases often originate from more than one individual and thus it is essential that typing systems can detect mixtures. Several studies are typically conducted to define the limitations of the DNA typing system. Genomic DNA from two samples of known genotype is often mixed in various ratios ranging from 20:1 to 1:20. The limit of detection for the minor component is determined by examining the profiles of the mock mixtures. Studies are also performed to examine the peak height ratios of heterozygote alleles within a locus and to determine the range of stutter percentages for each allele of each locus. The results of these relative peak height measurements can then be used to establish guidelines for separating a minor component of a mixture from the stutter product of a single-source sample.
- ***Precision and accuracy*** (2.9). The calculated base pair sizes for STR allele amplification products are measured. All measured alleles should fall within a 0.5 bp window around the measured size for the corresponding allele in the allelic ladder.
- ***PCR reaction conditions*** (2.10.1). Reaction conditions need to provide the required degree of specificity and robustness. Manufacturer developmental validation studies typically involve varying the amount or concentration of primers, DNA polymerase, magnesium chloride, and thermal cycling conditions.
  - *Annealing temperature studies*. These studies are commonly conducted by running the amplification protocol with the annealing temperature either two degrees above or two degrees below the optimal temperature. Annealing temperature studies are important because thermal cyclers might not always be calibrated accurately and can drift over time if not maintained properly. Thus, an operator might think that the annealing temperature during each cycle is 59 °C when in fact the thermal cycler is running hotter at 61 °C. If any primers in the multiplex mix are not capable of withstanding slight temperature variation (i.e., they do not hybridize as well), then a locus could drop out or non-specific amplification products could arise. While severe thermal cycler temperature drift is unlikely, understanding the limits of an assay or kit can help with troubleshooting.
  - *Cycle number studies.* The optimal PCR conditions (i.e., denaturing, annealing, and extension temperatures and times) are examined with a reduced number of cycles as well as a higher number of cycles than those called for by the standard protocol to evaluate the performance of the STR multiplex system. The sensitivity of detection of alleles for each locus is dependent of course on the quality of input DNA template.

The cycle number studies permit a laboratory to determine the tolerance levels of an STR multiplex system with various amounts of DNA template. While a higher number of PCR cycles (e.g., 34 instead of 28) might be able to better amplify very low levels of genomic DNA, the likelihood of non-specific amplification products arising increases with higher numbers of PCR cycles (see Chapter 11).

- ***PCR product characterization*** (2.10.5.1.1). Measurement standards for characterizing alleles need to be established. Common studies include examining peak height ratio variation in heterozygous samples and characterizing observed stutter percentages.
  - *Stutter studies.* The percentage of observed stutter at each STR locus is examined by calculating the ratio of the stutter peak area and/or peak height compared to the corresponding allele peak area and height. Stutter values are derived from homozygotes and heterozygotes with alleles separated by at least two repeat units. The upper levels of stutter observed for each locus are then used to develop interpretation guidelines. Because the levels of stutter for each of the commonly used tetranucleotide STR loci have been described and usually fall below 10% of the allele peak area and height, some labs just use a standard 15% cutoff for interpreting stutter products. If the stutter peak is below 15% of the allele peak, it is ignored as a biological artifact of the sample. However, if it is above 15% then a possible mixture could be present in the sample (Gill et al. 2006).
  - *Heterozygous peak height balance.* The peak heights of the smaller and the larger allele are compared typically by dividing the peak height of the smaller-sized allele height by the peak height of the larger-sized allele. In other words, the height of the lower peak in relative fluorescence units (RFU) is divided by the height of the higher peak (in RFU). This peak height ratio is expressed as a percentage. The average heterozygote peak height ratio is usually greater than 90% meaning that a heterozygous individual generally possesses well-balanced peaks. Ratios below 60% or 70% are rare in normal, unmixed samples (although sequence variation in the PCR primer binding region can cause one of the alleles to not amplify well).

The following studies are not commonly performed anymore because they have been shown to be true and thus repeating them would likely have little value:

- *Standard specimens* (4.1.5.1). DNA is isolated from different tissues and body fluids coming from the same individual and tested to make sure that the same type is observed. These studies were initially important to verify that blood samples or buccal swabs from a suspect could be used to match semen found at a crime scene.
- *Reproducibility* (4.1.5.4). Dried blood and semen stains are typed and compared to DNA profiles obtained from liquid samples. Samples from the same source should match. Obviously, this fact is important since a crime scene stain should match the reference blood sample of a suspect if he or she is the perpetrator of the crime.

As noted in QAS Standards 8.2.1 and 8.3.1 (QAS 2009), all validation studies should be documented. This documentation permits others to understand what studies have been performed in order to assess the extent of the validation work conducted. In some cases, hundreds or even thousands of samples may be tested as part of a developmental validation. These developmental validation experiments may also be coordinated across

multiple laboratories to spread the load as well as to demonstrate consistency between laboratories.

Table 7.4 contains a summary of the validation studies performed in the developmental validation of the PowerPlex Y kit (Krenke et al. 2004). This particular study involved 1269 samples examined as part of 17 different studies conducted by 8 laboratories.

TABLE 7.4 PowerPlex Y Developmental Validation Experiments (Distilled from Krenke et al. 2005).

Study Completed (17 studies performed)	**Description of Samples Tested (performed in 7 labs and Promega)**	**Number of Samples Analyzed**
Single source (Concordance)	5 samples × 8 labs	40
Mixture ratio (male:female)	6 labs × 2 M/F mixture series × 11 ratios (1:0, 1:1, 1:10, 1:100, 1:300, 1:1000, 0.5:300, 0.25:300, 0.125:300, 0.0625:300, 0.03:300 ng M:F)	132
Mixture ratio (male:male)	6 labs × 2 M/M mixtures series × 11 ratios (1:0, 19:1, 9:1, 5:1, 2:1, 1:1, 1:2, 1:5, 1:9, 1:19, 0:1)	132
Sensitivity	7 labs × 2 series × 6 amounts (1/0.5/0.25/0.125/0.06/0.03)	84
Non-human	24 animals	24
NIST SRM	6 components of SRM 2395	6
Precision(ABI 3100 and ABI 377)	10 ladder replicates + 10 samples replicated + [8 ladders + 8 samples for 377]	36
Non-probative Cases	65 cases with 102 samples	102
Stutter	412 males used	412
Peak Height Ratio	***N/A*** *(except for DYS385 but no studies were noted)*	–
Cycling parameters	5 cycles (28/27/26/25/24) × 8 punch sizes × 2 samples	80
Annealing temperature	5 labs × 5 temperatures (54/58/60/62/64) × 1 sample	25
Reaction volume	5 volumes (50/25/15/12.5/6.25) × [5 amounts + 5 concentrations]	50
Thermal cycler test	4 models (480/2400/9600/9700) × 1 sample + [3 models × 3 sets × 12 samples]	76
Male-specificity	2 females × 1 titration series (0-500 ng female DNA) × 5 amounts each	10
TaqGold DNA polymerase titration	5 amounts (1.38/2.06/2.75/3.44/4.13 U) × 4 quantities (1/0.5/0.25/0.13 ng DNA)	20
Primer pair titration	5 amounts (0.5x/0.75x/1x/1.5x/2x) × 4 quantities (1/0.5/0.25/0.13 ng DNA)	20
Magnesium titration	5 amounts (1/1.25/1.5/1.75/2 mM Mg) × 4 quantities (1/0.5/0.25/0.13 ng DNA)	20
	TOTAL SAMPLES EXAMINED	1269

Internal Validation of Established Procedures

In order to meet QAS requirements and to perform good science, forensic DNA laboratories conduct internal validation studies as part of becoming "validated." These studies demonstrate that DNA typing results can be consistently and accurately obtained *in the specific laboratory environment where the testing is performed*. Developmental validation demonstrates that the overall technology (i.e., STR kit or instrument) is reliable while internal validation demonstrates that a specific instrument or kit works properly in a specific laboratory environment. Internal validation anchors efforts within a laboratory to the previously performed developmental validation conducted by manufacturers or other laboratories.

Validation studies are performed with each new DNA typing system that is developed and used. For example, a lab may be validated with the Profiler Plus kit but it would need to perform additional validation studies when expanding its capabilities to amplifying the STR loci included in the Identifiler kit. The addition of a new instrument platform, such as the ABI 3500 (see Chapter 6), requires internal validation experiments to set detection and interpretation thresholds.

Typical studies for an internal validation include reproducibility, precision measurements for sizing alleles, and sensitivity (e.g., 50 ng down to 20 pg) along with mixture analysis and non-probative casework samples. In sizing precision studies, the calculated allele sizes in base pairs are plotted against the size deviation from the corresponding allele in the allelic ladder with which the genotype was determined. If a high degree of precision cannot be maintained due to laboratory conditions such as temperature fluctuations (see Chapter 6), then samples may not be able to be genotyped accurately.

In discussing internal validation, the 2004 SWGDAM Revised Validation Guidelines state: "The internal validation process should include the studies detailed below encompassing **a total of at least 50 samples**. Some studies may not be necessary due to the method itself" (SWGDAM 2004).

The seven sections from the 2004 SWGDAM guidelines that cover internal validation studies are listed below with some commentary and suggestions:

- ***Known and non-probative evidence samples*** (3.1). A method should first be evaluated using known samples where the quality and quantity of the sample itself is well-characterized and not a concern. In other words, the method being tested should be the variable not the sample being tested. Positive controls provided with STR typing kits, appropriate NIST Standard Reference Materials (see Table 7.7), or other well-characterized samples maintained in a laboratory can be used for the initial studies. Following the examination of known samples and demonstrating that the method is working properly, authentic or simulated case samples should be examined. The purpose of these samples is to explore the limits of a method in terms of handling typical case samples that may be degraded, contain PCR inhibitors, be limited in quantity, or contain a mixture of two or more individuals. The method must not only work on pristine DNA samples, but should also be able to cope with challenges present in forensic specimens. Non-probative evidence samples, such as biological material from a previously completed case, may be used if sufficient quantities are available. Alternatively, artificial case-similar samples may be created (e.g., artificial mixtures as described in the mixture studies section below).

- ***Reproducibility and precision*** (3.2). The laboratory needs to document a method's reproducibility and precision using appropriate control samples that are repeatedly tested. Replicate injections of an allelic ladder enables sizing precision to be determined for each allele spanning the size range of the STR typing kit. Typically, 5 to 10 injections (see Figure 7.1) of an allelic ladder will provide adequate information on STR allele sizing reproducibility.
- ***Match criteria*** (3.3). For procedures that entail separation of DNA molecules based on size, precision of sizing should be determined by repetitive analyses of appropriate samples to establish criteria for matching or allele designation. Typically genotyping software utilizes a ±0.5 bp window so that alleles that are 1 bp apart can be appropriately designated. Match criteria depend on results from the precision experiments. A simple way to test match criteria when validating STR typing kits is to run more than one allelic ladder with a batch of samples—but only designate the first allelic ladder in the analysis software as a "ladder" for calibration purposes. If all of the alleles present in the other ladders, which are treated as "samples" rather than "ladders," are designated correctly, then the match criteria used by the analysis software is working appropriately. If on the other hand, some of the ladder alleles are designated "off-ladder," then room temperature may be shifting during the course of the sample batch processing time (see Chapter 6). Environmental conditions would have to be changed to restore run-to-run precision. Alternatively, procedural changes may be required to ensure reliable results, such as running an allelic ladder for calibration purposes on a more frequent basis.
- ***Sensitivity and stochastic studies*** (3.4). Reliability and integrity of sample results are important. PCR-based methods are very sensitive but can be subject to stochastic sampling variation when low amounts of DNA are being amplified (see Chapter 11). Limits of sensitivity are typically characterized through testing a dilution series of one or more samples. Upper limits of performance with a method should also be studied and understood. Observing the impact of adding both too much and too little DNA to a PCR amplification can be instructive in defining the optimal range and limits of the method being validated. Thus, a sensitivity study for validating a new STR typing kit may include amounts of 5 ng, 2 ng, 1 ng, 0.5 ng, 0.25 ng, 0.1 ng, 0.05 ng, 0.03 ng, and 0.01 ng from one or more DNA samples when the optimital template target range is 0.5 ng to 1 ng.
- ***Mixture studies*** (3.5). Forensic casework laboratories should define and mimic the range of detectable mixture ratios. The ability to reliably distinguish between major and minor components in a mixture should also be examined as part of mixture studies. For example, artificial mixtures may be created by combining one part female DNA with a serial dilution of male DNA and vice versa. Mixture ratios studied may include 1:1, 1:2, 1:3, etc., in increments down to 1:10 or more. Samples that mimic casework are also valuable to use in these studies such as postcoital vaginal swabs that contain a mixture of male and female cells.
- ***Contamination*** (3.6). The laboratory needs to demonstrate that the method under investigation, as well as its general procedures, minimizes contamination that would compromise the integrity of results obtained. Negative PCR amplification controls and extraction blanks are typically used to assess the purity of the reagents used. For example, if fluorescent dye artifacts are present in a specific batch of PCR primers, then they will be observed in the negative control.

- ***Qualifying test*** (3.7). Analysts who are becoming qualified to perform a specific method need to demonstrate that they are proficient with the method. This qualifying test may involve using previous proficiency samples or other types of samples routinely examined by the laboratory. For example, a previously characterized bone sample could be used by a mitochondrial DNA laboratory as a qualifying test for a new analyst. Once the analyst becomes qualified then he or she enters the cycle of proficiency testing required by the laboratory to demonstrate that competenancy is being maintained over time.

Additional Efforts with Bringing a Method "Online"

Besides the actual internal validation experiments that are performed, there is additional effort typically required to bring a new method "online" before it can be used in forensic casework. Some of this effort takes place before the validation experiments and some after. There is pre-validation learning and preparation and post-validation training.

In their 2006 "Validation and Implementation of (New) Methods" guidance document, ENFSI notes that the level of validation required depends on the type of method being validated (ENFSI 2006). ENFSI establishes categories of (1) an existing standard method, (2) a changed method, and (3) a new method, which is often called an "in-house" method.

Bringing a procedure (assay, instrument, or software) "online" in a forensic lab setting typically includes (1) learning about the technique and how to perform it properly through studying the literature and talking to others who have previously implemented the technique, (2) installing the instrumentation or software and purchasing the assay reagents, (3) designing a validation plan with appropriate experiments to assess reliability and range (limitations) of the procedure, (4) assembling a set of samples for testing based on the validation plan, (5) examining the analytical procedure through conducting predetermined validation studies and maintaining documentation on results, (6) summarizing the studies and getting approval of the laboratory's technical leader, (7) creating standard operating procedures for the laboratory (including controls and critical reagents to be used for quality assurance purposes) with interpretation guidelines based on the validation studies, and (8) training other personnel on the procedure. Each trained analyst will then need to pass a qualifying test before he or she can use the procedure in forensic casework.

After a procedure has been successfully implemented for use with forensic casework, proficiency tests are performed on a regular basis (usually twice a year) to demonstrate successful application of the technique over time by qualified analysts. In addition, new materials and instruments need to be evaluated over time through a quality control process involving a performance check on the validated procedure.

Designing an Internal Validation Study

While the 2004 SWGDAM guidelines provide helpful information, they do not include specific recommendations in terms of the minimum number of samples to be tested for each internal validation study. The only concrete figure provided in the SWGDAM guidelines is that a minimum of at least 50 samples should be run for internal validation purposes by a forensic laboratory.

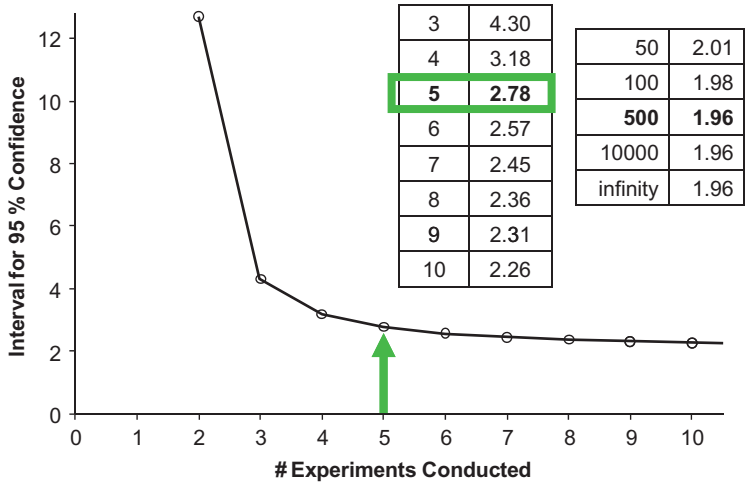


FIGURE 7.1 Student's *t* distribution showing the impact of the number of experiments on capturing variability in a population of data. Five replicate experiments works well as a minimum sample number because there are diminishing returns with each successive experiment.

While everyone may have different comfort levels and absolute numbers for defined studies will not likely be widely accepted, it is worth noting that each study does not require an excessive number of samples to be run and that not all validation studies are essential for every procedure under consideration. Validation should not be an attempt to mimic every potential situation that may be seen in future casework within a laboratory. Rather, data collected during validation should relate to how well potential data to be collected during future work using a specific method should perform.

The Student's *t* distribution is a statistical approach to associate how well a sample (or set of data) can estimate the characteristics of a population of data (or larger set of data). With each additional repeated experiment, greater confidence is typically gained in the measured result. Figure 7.1 shows a Student's *t* curve with the 95% confidence interval decreasing as the number of replicate experiments conducted increases. Note that there are diminishing returns from repeating more than 5 to 10 replicates in terms of the confidence around a set of data (Figure 7.1). This figure illustrates why five replicate experiments are often selected as a minimum sample number.

There is no requirement for how long a validation study should take and thus the "at least 50 samples" recommended by SWGDAM could be run in a matter of days rather than weeks or months. Not all studies may be necessary due to the method involved (D.N.A. Box 7.2). Experimental design can also be implemented in many studies to aid in examining the variables under investigation. Some suggested approaches involving experimental design are available at http://www.haag.com/Seminars_files/How%20Many%20Data%20Handout.pdf.

Companies supplying commercial STR kits have created validation guides to aid laboratories in developing and conducting internal validation studies. Validation services are also available where a company will for a fee come into a forensic laboratory, set up a new

D.N.A. BOX 7.2

REVIEW OF URBAN LEGENDS

In September 2006 I published an article reviewing eight 'urban legends' surrounding validation (Butler 2006). The urban legends discussed in this article included the following:

1. Hundreds or thousands of samples are required to fully validate an instrument or method.
2. Validation is uniformly performed throughout the community.
3. Each component of a DNA test or process must be validated separately.
4. Validation should seek to understand everything that could potentially go wrong with an instrument or technique.
5. Learning the technique and training other analysts are part of validation.
6. Validation is boring and should be performed by summer interns since it is beneath the dignity of a qualified analyst.
7. Documenting validation is difficult and should be extensive.
8. Once a validation study is completed you never have to revisit it.

As technology advances and new methods are developed, there will always be something to validate in a laboratory. A primary purpose is writing the *Urban Legends* article was to help analysts appreciate that validation requires common sense and is best performed (where possible) with well-characterized samples through concordance to results produced from previous methods. Some aspects of validation can be achieved with a minimal amount of DNA samples while other aspects will require more extensive studies. In November 2010, the European Network of Forensic Science Institutes (ENFSI) DNA Working Group QA/QC subcommittee released a document building on the *Urban Legends* article and provided more detail on various aspects of the DNA typing process (ENFSI 2010).

Treating validation as a one-time event that is performed by a single individual (perhaps a summer intern who leaves the lab after performing the measurements) can lead to problems. Every analyst that is interpreting DNA typing data should be familiar with and understand the validation studies that hopefully underpin the laboratory's standard operating procedures. Validation defines the scope of a technique and thus its limitations. Making measurements around the edges of what works well will help better define the reliable boundaries of the technique. While developmental validation may be broadly applicable, internal validation is not transferrable in the same way. The performance characteristics and limitations of an instrument, a software program, and a DNA typing assay are important to understand in order to effectively interpret forensic DNA data.

Sources:

Butler, J.M. (2006). Debunking some urban legends surrounding validation within the forensic DNA community. Profiles in DNA, 9(2), *3-6. Available at http://www.promega.com/profiles/.*

ENFSI DNA Working Group (2010). Recommended minimum criteria for the validation of various aspects of the DNA profiling process. Available at http://www.enfsi.eu.

instrument or assay, perform validation experiments, produce validation documentation, and train analysts on the new instrument or assay.

Publication and Presentation of Validation Results

Results of developmental validation studies are shared as soon as possible with the scientific community either through presentations at professional meetings or publication in peer-reviewed journals. Rapid dissemination of information about these studies is important to the legal system that forensic science serves because the courts must decide whether or not the DNA evidence is admissible (see Chapter 18). Publication of a technology or methodology in a peer-reviewed journal is one method of showing that it is generally accepted in the scientific field. Thus, revised QAS Standard 8.2.2 requires peer-reviewed publication of the underlying scientific principle(s) of a technology (QAS 2009). Table 7.5 contains a listing of some published developmental and internal validation studies.

The most commonly used scientific journals for publishing validation studies and population data results are the *Journal of Forensic Sciences (JFS)*, the *International Journal of Legal Medicine*, *Legal Medicine*, and *Forensic Science International (FSI)* (and since 2007 its daughter journal *FSI Genetics*). Other forensic journals include *Science and Justice*, *Canadian Society of Forensic Science Journal*, *American Journal of Forensic Medicine and Pathology*, *Forensic Science Review*, and *Forensic Science, Medicine, and Pathology*. Forensic DNA studies have also been published in *Analytical Chemistry*, *Electrophoresis*, *Human Mutation*, *Genome Research*, and other scientific journals. The FBI Laboratory maintains an online journal entitled *Forensic Science Communications* that is published quarterly and is available at http://www2.fbi.gov/hq/lab/fsc/current/index.htm. Prior to 1999, this journal was known as *Crime Laboratory Digest*.

Scientific meetings where DNA typing research and validation studies are presented include the *International Symposium on Human Identification* (ISHI, sponsored each fall by the Promega Corporation), the *American Academy of Forensic Sciences* (AAFS, held each February), the *Congress of the International Society of Forensic Genetics* (ISFG, held in late summer biannually), and the *International Association of Forensic Sciences* (IAFS, held every three years in the summer months). An examination of the first 14 years of the ISHI meeting abstracts found greater than 10% of the 1220 presentations covered validation studies based on the title of the talk or poster (Butler et al. 2004).

Within the United States, there are several additional meetings sponsored by private companies involved in the forensic DNA community. Each spring Bode Technology Group sponsors an East Coast conference and a West Coast conference. During the summer months, Applied Biosystems and Promega Corporation often conduct traveling road shows to discuss their latest products.

Several regional U.S. forensic organizations hold annual meetings including the Northeastern Association of Forensic Scientists (NEAFS), the Mid-Atlantic Association of Forensic Scientists (MAAFS), the Midwestern Association of Forensic Scientists (MAFS), the Northwestern Association of Forensic Scientists (NWAFS), the Southwestern Association of Forensic Scientists (SWAFS), the Southern Association of Forensic Scientists (SAFS), and the California Association of Criminalists (CAC). These meetings often have workshops associated with them where validation data are shared.

TABLE 7.5 Published Validation Studies Conducted Using Various STR Kits, Assays, Instruments, or Software.

Kit, Assay, or Instrument	Reference (D = developmental validation; I = internal validation)
AmpFlSTR Blue	**D** – Wallin et al. (1998)
AmpFlSTR Green I	**D** – Holt et al. (2002)
COfiler	**D** – Holt et al. (2002), Wallin et al. (2002); **I** – LaFountain et al. (2001), Moretti et al. (2001), Tomsey et al. (2001), Buse et al. (2003)
Identifiler	**D** – Collins et al. (2004)
Identifiler (low template)	**D** – Caragine et al. (2009)
MiniFiler	**D** – Mulero et al. (2008), **I** – Luce et al. (2009)
Profiler	**D** – Holt et al. (2002)
Profiler Plus	**D** – Holt et al. (2002), Wallin et al. (2002), Frégeau et al. (2003); **I** – Pawlowski & Maciejewska (2000), Frank et al. (2001), LaFountain et al. (2001), Moretti et al. (2001), Tomsey et al. (2001), Buse et al. (2003)
Profiler Plus (reduced volume)	**D** – Frégeau et al. (2003); **I** – Gaines et al. (2002)
Profiler Plus *ID*	**D** – Leibelt et al. (2003)
SEfiler	**D** – Coticone et al. (2004)
SGM Plus	**D** – Cotton et al. (2000)
Yfiler	**D –** Mulero et al. (2006); **I** – Gross et al. (2008)
CTT	**D** – Budowle et al. (1997)
PowerPlex 1.1	**D** – Micka et al. (1999); **I** – Tomsey et al. (2001), Greenspoon et al. (2001)
PowerPlex 1.1 + D16 primer	**I** – Nelson et al. (2002)
PowerPlex 2.1	**D** – Levedakou et al. (2002); **I** – Tomsey et al. (2001)
PowerPlex 16	**D** – Krenke et al. (2002); **I** – Tomsey et al. (2001)
PowerPlex 16 (reduced volume)	**I** – Spathis & Lum (2008)
PowerPlex 16 BIO	**D** – Greenspoon et al. (2004b)
PowerPlex 16 HS	**D** – Ensenberger et al. (2010)
PowerPlex Y	**D** – Krenke et al. (2005)
PowerPlex ESI 16 & ESI 17	**D** – Tucker et al. (2011)
Y-PLEX 6	**D** – Sinha et al. (2003a)
Y-PLEX 5	**D** – Sinha et al. (2003b)
Y-PLEX 12	**D** – Shewale et al. (2004)
genRES MPX-2	**D** – Junge et al. (2003)
Amelogenin	**I** – LaFountain et al. (1998)
D3S1358, D8S1179, D18S51	**I** – Potter (2003)

(Continued)

TABLE 7.5 Published Validation Studies Conducted Using Various STR Kits, Assays, Instruments, or Software. (*Continued*)

Kit, Assay, or Instrument	Reference (D = developmental validation; I = internal validation)
D12S391	**D** – Junge et al. (1999)
TH01	**D** – van Oorschot et al. (1996), Wiegand et al. (1993)
TH01, VWA, F13A1, FES, LPL	**D** – Pestoni et al. (1995)
TH01, VWA, F13A1, FES	**D** – Kimpton et al. (1994); **I** – Lygo et al. (1994), Clayton et al. (1995), Andersen et al. (1996)
SGM	**D** – Sparkes et al. (1996a), Sparkes et al. (1996b), Kimpton et al. (1996)
STR sets	**D** – Crouse & Schumm (1995), Micka et al. (1996)
Q8 (short amplicon multiplex)	**D** – Muller et al. (2009)
Miniplexes (short amplicons)	**D** – Opel et al. (2007)
Y-STR 4plex	**D** – Prinz et al. (2001)
Y-STR 10plex	**D** – Johnson et al. (2003)
Y-STR 19plex	**D** – Daniels et al. (2004)
X-STR 12plex	**D** – Turrina et al. (2007)
Eurasian badger STRs	**D** – Dawnay et al. (2008)
Canine STRs	**D** – Dayton et al. (2009)
Marijuana STRs	**D** – Howard et al. (2008)
RSID – blood	**D** – Schweers et al. (2008)
RSID – saliva	**D** – Old et al. (2009)
qPCR *Alu* DNA quant	**D** – Nicklas & Buel (2003)
qPCR CFS TH01 DNA quant	**D** – Richard et al. (2003)
qPCR CA DOJ quadruplex	**D** – Hudlow et al. (2008)
AluQuant DNA quant assay	**D** – Mandrekar et al. (2001)
BodeQuant	**D** – Fox et al. (2003)
Plexor HY	**D** – Krenke et al. (2008)
Quantifiler	**D** – Green et al. (2005)
Quantifiler (reduced volume)	**I** – Westring et al. (2007)
Quantifiler Duo	**D** – Barbisin et al. (2009)
Feline, bovine, equine, & cervid qPCR assays	**D** – Lindquist et al. (2011)
mtDNA sequencing	**D** – Wilson et al. (1995), Holland & Parsons (1999); **I** – Jarman et al. (2009)
mtDNA minisequencing	**D** – Morley et al. (1999)
Biomek 2000 with DNA IQ	**D** – Greenspoon et al. (2004a)

(Continued)

TABLE 7.5 Published Validation Studies Conducted Using Various STR Kits, Assays, Instruments, or Software. (*Continued*)

Kit, Assay, or Instrument	Reference (D = developmental validation; I = internal validation)
PrepFiler	**D** – Brevnov et al. (2009)
SNP autosomal 21plex	**D** – Dixon et al. (2005)
SNP*for*ID 52plex	**D** – Musgrave-Brown et al. (2007)
IrisPlex	**D** – Walsh et al. (2011)
ABI 377	**D** – Frazier et al. (1996), Fregeau et al. (1999)
ABI 310	**D** – Lazaruk et al. (1998); **I** – Isenberg et al. (1998), Moretti et al. (2001b)
ABI 3100	**D** – Koumi et al. (2004); **I** – Sgueglia et al. (2003)
ABI 3700	**D** – Gill et al. (2001), Koumi et al. (2004)
ABI 3730xl	**D** – Spathis & Lum (2008)
MegaBACE	**D** – Koumi et al. (2004)
TrueAllele software	**D** – Kadash et al. (2004)
CompareCalls software	**I** – Ryan et al. (2004)

Documenting and Applying Validation Information

Some journals view validation results as not sufficiently novel to warrant publication (Buckleton 2009). Yet information within the validation studies is important and should be shared with the community. Early adopters of a new technology or DNA testing kit should be able to share their experiences with future users. If utilized as a hub for sharing information, websites, such as the NIST STRBase validation section discussed below, could benefit the forensic DNA community around the world. Such posting of information from individual forensic laboratories' internal validation studies can help the community quickly gain a sense of the numbers of samples and types of samples run in other labs.

The NIST STRBase website has been widely used by the forensic DNA typing community since it was introduced in 1997. A validation section was created in 2004 that can serve as a repository of helpful information on kit, assay, software, and instrument validation studies performed in forensic DNA laboratories (see http://www.cstl.nist.gov/biotech/strbase/validation.htm). This site also contains links to the FBI Quality Assurance Standards and the SWGDAM Revised Validation Guidelines. Publications in the literature involving validation of forensic DNA tests are also listed and some are summarized with Validation Summary Sheets like the information shown in Table 7.4 for the developmental validation of PowerPlex Y (Krenke et al. 2005).

OTHER AIDS TO QUALITY ASSURANCE

There are several other aids to quality assurance that can strengthen confidence in results. These include interlaboratory tests, certified reference materials, and commercial STR kits.

Automation of data interpretation as well as sample handling and tracking (see Chapter 17) also help production of high-quality DNA typing results.

Exploratory Interlaboratory Tests

Exploratory interlaboratory tests are one way that the forensic community uses to demonstrate that multiple laboratories can generate comparable results with the same DNA samples. Since DNA databases (see Chapter 8) rely on information contributed from multiple laboratories, interlaboratory studies that demonstrate consistency in results across laboratories on the same DNA samples can increase confidence in work performed by these laboratories. The reference listing at the end of this chapter includes 53 interlaboratory publications in the past two decades.

Since 1994, the European DNA Profiling Group (EDNAP) has conducted a series of interlaboratory evaluations on various STR loci and methodologies used for analyzing them. Each study involved the examination of five to seven bloodstains that were distributed to multiple laboratories (usually a dozen or more) to test their abilities to obtain consistent results. In all cases where simple STR loci were tested, consistent results were obtained. However, in early studies, complex STR markers, such as ACTBP2 (SE33), often gave inconsistent results. Thus, at that time, STRs with complex repeat structures were not recommended for use in DNA databases in which results are submitted from multiple laboratories. The availability of commercial kits and allelic ladders that enable consistent amplification and typing of SE33 now means that obtaining reproducible results is more feasible.

In the United States several interlaboratory studies have been performed. The first large test with commercial STR typing kits was conducted by the National Institute of Standards and Technology (NIST) and involved 34 laboratories that evaluated the three STRs TH01, TPOX, and CSF1PO in a multiplex amplification format (Kline et al. 1997). This study concluded that as long as locus-specific allelic ladders were used, a variety of separation and detection methods could be used to obtain equivalent genotypes for the same samples.

NIST has also conducted several DNA quantitation and mixture interlaboratory studies (Duewer et al. 2001, Kline et al. 2003, Kline et al. 2005). While different instrument sensitivities have been observed between laboratories, these studies have generally demonstrated that consistent results can be obtained between participating laboratories, thus helping support the conclusion that forensic DNA typing methods are reliable and reproducible when performed properly.

Table 7.6 contains a summary of NIST and EDNAP interlaboratory studies conducted since the early 1990s. These studies have enabled laboratories to learn from one another, to improve their techniques, and to implement new methodologies with greater confidence. The reference list at the back of the chapter includes citations to publications involving 20 EDNAP and 15 Spanish and Portugese-Speaking Working Group of the ISFG (GEP-ISFG) collaborative exercises along with 13 publications describing NIST-coordinated interlaboratory studies.

Since 1991, the English-Speaking Working Group of the ISFG has conducted paternity testing workshops. Each year a paper challenge is provided to requesting laboratories in order to assess reporting results and calculations used in relationship testing. A series of publications

TABLE 7.6 Summary of NIST and EDNAP Interlaboratory Collaborative Studies.

Study/Purpose	# Labs	References
NIST		
RFLP studies	22	Mudd et al. (1994)
	20	Duewer et al. (1995)
	22	Stolorow et al. (1996)
	22	Duewer et al. (1997)
	36	Duewer et al. (1998)
	29/20	Duewer et al. (2000)
	51 (>1000 data sets)	Duewer et al. (2000)
Evaluation of CSF1PO, TPOX, and TH01	34	Kline et al. (1997)
Mixed Stain Studies #1 and #2 (Apr.–Nov 1997 & Jan–May 1999)	45	Duewer et al. (2001)
Mixed Stain Study #3 (Oct 2000-May 2001)	74	Kline et al. (2003) Duewer et al. (2004)
DNA Quantitation Study (Jan-Mar 2004)	80	Kline et al. (2005)
Mixture Interpretation Study (Jan-June 2005)	69	Kline & Butler (2005)
EDNAP		
RFLP studies: single locus VNTR probe (different protocols)	11	Schneider et al. (1991)
RFLP studies: uniformity obtainable (same protocol)	14	Gill et al. (1992)
Examination of TH01 and ACTBP2 (SE33)	14	Gill et al. (1994)
Examination of TH01, VWA, FES/FPS, F13A1	30	Kimpton et al. (1995)
Evaluation of TH01 and VWA	16	Andersen et al. (1996)
Evaluation of D21S11 and FGA	16	Gill et al. (1997)
Examination of ACTBP2, APAI1 and D11S554; Evaluation of D12S391 in 7 labs and D1S1656	7; 12	Gill et al. (1998)
Evaluation of mtDNA sequencing	12	Carracedo et al. (1998)
Use of DYS385	14	Schneider et al. (1999)
Evaluation of pentaplex for DYS19, DYS389I/II, DYS390, and DYS393	18	Carracedo et al. (2001)
Examination of artificially degraded DNA samples	38	Schneider et al. (2004)
Examination of mtDNA sequencing issues	21	Parson et al. (2004)
Evaluation of mtDNA heteroplasmy in hairs	10	Tully et al. (2004)

(Continued)

TABLE 7.6 Summary of NIST and EDNAP Interlaboratory Collaborative Studies. (*Continued*)

Study/Purpose	# Labs	References
Evaluation of 11 Y-SNP markers (SRY-1532, M40, M35, M213, M9, 92R7, M17, P25, M18, M153, M167)	8	Brion et al. (2005)
Evaluation of degraded DNA with low-copy number STR profiling, miniSTRs, and SNPs	9	Dixon et al. (2006)
West Eurasian mitochondrial haplogroups by mtDNA SNP screening	12	Parson et al. (2008)
Typing of autosomal SNPs with a 29 SNP-multiplex	12	Sanchez et al. (2008)
Comparison of miniSTRs (MiniFiler) versus standard STRs (SGM Plus)	4	Welch et al. (2011)
mRNA profiling for the identification of blood	16	Haas et al. (2011)
SNP typing with GenPlex HID 48plex	14	Tomas et al. (2011)

have described the results of these interlaboratory studies for 1991 to1994 (Syndercombe-Court & Lincoln 1995), 1995 and 1996 (Bjerre et al. 1997), 1997 to 1999 (Hallenberg & Morling 2001), 2000 and 2001 (Hallenberg & Morling 2002), and 2002 to 2008 (Thomsen et al. 2009).

Certified Reference Materials

One of the primary ways to support a consistent and calibrated STR allele nomenclature is for DNA testing laboratories to use common reference materials. The National Institute of Standards and Technology, which is a non-regulatory agency in the U.S. Department of Commerce, provides reference materials for a variety of fields to enable accurate and comparable measurements. NIST supplies over 1300 reference materials to industry, academia, and government laboratories to facilitate quality assurance and support measurement traceability. These Standard Reference Materials (SRMs) are certified through carefully characterizing the properties for which values are assigned. The term SRM is the NIST name for a certified reference material.

Reference DNA samples are crucial to the validation of any DNA testing procedure. Standard 9.5.5 in the revised Quality Assurance Standards states: "The laboratory shall check its DNA procedures annually or whenever substantial changes are made to the protocol(s) against an appropriate and available NIST standard reference material or standard traceable to a NIST standard."

NIST supplies several DNA SRMs to enable validation of a laboratory's measurement capabilities as well as calibration of instrumentation and methods (Table 7.7). Current SRMs used by the forensic DNA community include SRM 2391b PCR-Based DNA Profiling Standard for autosomal STR markers, SRM 2392-I Mitochondrial DNA Sequencing for mtDNA sequence information, SRM 2395 Human Y-Chromosome DNA Profiling Standard for Y-chromosome markers, and SRM 2372 Human DNA Quantitation Standard for human DNA quantitation.

TABLE 7.7 NIST SRMs for Use in Forensic DNA or Human Identity Testing Applications (as of 2010).

NIST Standard Reference Material	Release date and renewals	Purpose/contents
SRM 2390	1992	DNA profiling standard (RFLP)
SRM 2391b	1995, 1999, 2002, 2008	PCR-based DNA profiling standard (STRs)
SRM 2392-I	2003, 2009	Mitochondrial DNA sequencing (human HL-60 DNA)
SRM 2395	2003, 2008	Human Y-chromosome DNA profiling standard
SRM 2372	2007	Human genomic DNA quantitation standard

A review of the various SRM materials that are now or have been available to aid the forensic DNA typing community is given below.

RFLP Testing Standard: NIST SRM 2390

SRM 2390 DNA Profiling Standard (RFLP-based typing methods) was released in August 1992 and for many years was used in standardizing forensic and paternity testing quality assurance procedures for restriction fragment length polymorphism (RFLP) testing that used *Hae*III restriction enzymes as well as instructional law enforcement and non-clinical research purposes. It contained two well-characterized human DNA samples: a female cell line (K562) and a male source (TAW). Both samples were available in three forms: as a cell pellet (3×10^6 cells), an extracted genomic DNA (≈200 ng/μL), and a *Hae*III restriction digest (pre-cut DNA; 25 ng/μL). A relative molecular mass marker for DNA sizing purposes and six quantitation standards (250 ng, 100 ng, 50 ng, 25 ng, 12.5 ng, and 6 ng) were also included along with agarose for slab gel preparation. Certified values for the DNA band sizes were available for five commonly used RFLP markers. These markers (and variable number of tandem repeat probes) were D2S44 (YNH24), D4S139 (PH30), D10S28 (TBQ7), D1S7 (MS1), and D17S79 (V1). The certified values represented the pooled results from analyses performed at NIST and 28 collaborating laboratories and came with calculated uncertainties (see Duewer et al. 2000). With the change in technology from RFLP to PCR in the mid-1990s, SRM 2390 is no longer used.

Autosomal STR Testing Standard: NIST SRM 2391b

SRM 2391b PCR-Based DNA Profiling Standard was reissued in 2002. It is an update of the original SRM 2391 that became available in 1995 and includes certified values for new STR loci. SRM 2391b is intended for use in standardizing forensic and paternity testing quality assurance procedures involving polymerase chain reaction (PCR)-based genetic testing as well as instructional law enforcement and non-clinical research purposes.

It contains 12 components of well-characterized DNA in two forms: genomic DNA and DNA to be extracted from cells spotted on filter paper. There are 10 genomic DNA samples,

all at a concentration of 1 ng/μL (20 μL volume). Cell lines 9947A and 9948 are included on a 6 mm Schleicher & Schuell 903 filter paper circle spotted with 5×10^4 cells. The cells permit a laboratory to test its ability to perform DNA extraction while the genomic DNA materials may be used to verify reliable PCR amplification and detection technologies.

Certified genotype values for the 12 SRM components are listed for the FBI's CODIS 13 STR loci (CSF1PO, D3S1358, D5S818, D7S820, D8S1179, D13S317, D16S539, D18S51, D21S11, FGA, TH01, TPOX, and VWA) as well as additional STR loci F13A01, F13B, FES/FPS, LPL, Penta D, Penta E, D2S1338, D19S433, and SE33. These STR markers are all available in commercial kits from either the Promega Corporation or Applied Biosystems. Certified values for the genetic loci HLA-DQA1, PolyMarker, D1S80, and amelogenin were removed for the 12 components in the last update as these kits are no longer commercially available or in use by the forensic community. In 2008, certified values were added for 26 miniSTR loci (see Chapter 10). When the supply of SRM 2391b runs out, it will be replaced by SRM 2391c.

Mitochondrial DNA Testing Standard: NIST SRM 2392-I

SRM 2392-I Mitochondrial DNA Sequencing (Human) Standard was released in June 2003. It was recertified in 2009. This SRM is intended to provide quality control when performing the polymerase chain reaction (PCR) and sequencing human mitochondrial DNA (mtDNA) for forensic investigations, medical diagnosis, or mutation detection as well as to serve as a control when PCR-amplifying and sequencing any DNA sample.

SRM 2392-I contains extracted DNA from the human cell line HL-60 (65 μL DNA at 1.4 ng/μL) that has been sequenced across the entire mtDNA genome (Levin et al. 2003). A list of 58 unique primer sets that were designed to amplify any portion or the entire human mtDNA genome is also included.

Human Y-Chromosome DNA Profiling Standard: NIST SRM 2395

SRM 2395 Human Y-Chromosome DNA Profiling Standard was released in July 2003 for use with verifying results involving Y-chromosome STR testing (see Chapter 13). SRM 2395 includes five male DNA samples (50 μL at 2 ng/μL) selected to exhibit a diverse set of alleles across 31 commonly used Y chromosome STRs and 42 single nucleotide polymorphism (SNP) markers. A female DNA sample is also included to serve as a negative control for male-specific DNA tests. In addition to the typing results from all commercially available Y-STR kits, the five male samples in SRM 2395 have been sequenced at 22 Y-STR loci to confirm allele calls (Kline et al. 2003b). SRM 2395 was recertified in 2008, and information was added on 20 additional Y-STRs useful for genetic genealogy testing.

Human DNA Quantitation Standard: NIST SRM 2372

SRM 2372 Human DNA Quantitation Standard was released in October 2007 to aid DNA quantitation (see Chapter 3). It contains three components: a single-source male DNA sample, a multiple-source female DNA sample, and a DNA sample containing a mixture of male

and female sources. Each of these components was certified for decadic attenuance (absorbance) at 260 nm with informational values provided based on the conventional conversion that an optical density of 1.0 at 260 nm in a cuvette with a 1 cm pathlength has a DNA concentration of 50 ng/μL (Kline et al. 2009).

Commercial STR Kits and Concordance Studies

In the early days of STR typing, forensic laboratories put together their own PCR mixes, primer sets, and allelic ladders. This meant that variation existed in the materials used for various laboratories and sometimes in the interpretation of a sample's genotype. Laboratories often had to spend a significant amount of time preparing the allelic ladders and verifying that each lot of primer mix worked appropriately.

Today, commercially available STR typing kits (see Chapter 5) help to maintain a high level of confidence in results and to ensure consistency in nomenclature between laboratories. Use of commercial kits does increase the cost of DNA testing but aids in overall quality assurance due to compatibility and consistency of results (both in terms of loci examined and STR allele nomenclature used). These commercial kits come with company-supplied allelic ladders, which are composed of common alleles and used in sample data interpretation to make the specific STR allele designations.

Production of STR kits by commercial manufacturers requires extensive quality control. A fluorescent dye is attached to one primer for each locus amplified by the multiplex STR kit. Each primer must be purified and combined in the correct amount in order to produce a balanced amplification. Variation in this primer mix production can affect locus-to-locus balance in the multiplex amplification. In addition, allelic ladders must also be produced on a large scale and be well-characterized since they serve as the standard for performing the DNA typing experiments with unknown samples.

As discussed in Chapter 8, DNA databases may acquire DNA profiles from STR typing kits that utilize different PCR primers. Concordance studies, where the same DNA samples are tested with different STR kits, are important to locate potential primer binding site mutations that could lead to allele drop-out and discordance of results (Figure 7.2). Fortunately, concordance studies have shown that equivalent results can be obtained greater than 99.9% of the time (Hill et al. 2010). A null allele page is available on the NIST STRBase website to catalog observed differences between kits: http://www.cstl.nist.gov/biotech/strbase/NullAlleles.htm.

Summary and Final Thoughts

Quality assurance of the DNA typing process requires diligence and action at many levels to help maintain reliable scientific results. A number of organizations play crucial roles in standardizing efforts internationally so that DNA data can be shared among laboratories. Internal validation experiments need to be carefully performed and should serve as the basis for interpretation protocols, thresholds, and decisions. Much can be learned from interlaboratory studies or collaborative exercises where multiple laboratories run the same samples and compile results obtained.

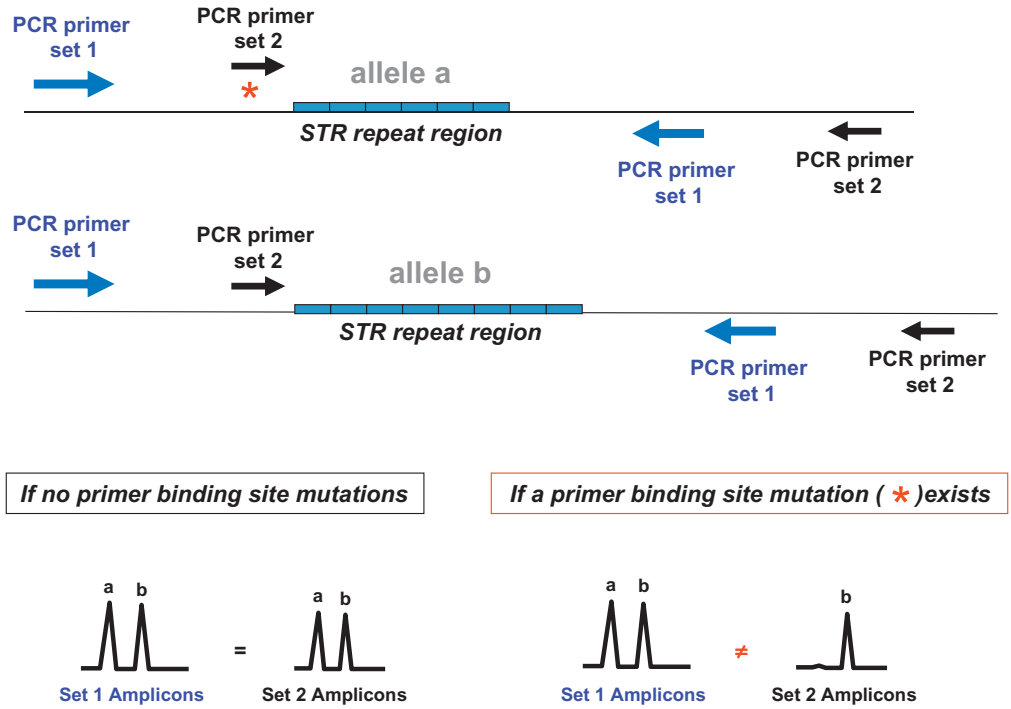


FIGURE 7.2 When different PCR primer sets are utilized to amplify the same genetic marker, sequence variability in the primer binding regions (represented as an asterisk upstream of the STR repeat region in allele a) could cause one of the primers to not fully anneal to the DNA template. In this case, PCR could fail to amplify this DNA target and result in allele drop-out that reproducibly impacts one set of primers but not another.

READING LIST AND INTERNET RESOURCES

Quality Assurance

Aboul-Enein, H. Y., et al. (2001). *Quality and reliability in analytical chemistry*. Washington, DC: CRC Press.

Cormier, K. L., et al. (2005). Evolution of the quality assurance documents for DNA laboratories. *Forensic Magazine*, *2*(1), 16–19.

Christian, G. D. (2004). *Analytical chemistry* (6th ed.). New York: John Wiley & Sons.

ISO/IEC 17025:2005 (2005). *General requirements for the competence of testing and calibration laboratories*. Available from <http://www.iso.org>.

Malkoc, E., & Neuteboom, W. (2007). The current status of forensic science laboratory accreditation in Europe. *Forensic Science International*, *167*, 121–126.

Presley, L. A. (1999). The evolution of quality assurance standards for forensic DNA analyses in the United States. *Profiles in DNA*, *3*(2), 10–11.

Roper, P., et al. (2001). *Applications of reference materials in analytical chemistry*. Cambridge, UK: Royal Society of Chemistry.

Reeder, D. J. (1999). Impact of DNA typing on standards and practice in the forensic community. *Archives of Pathology & Laboratory Medicine*, *123*, 1063–1065.

Schneider, P. M. (2007). Scientific standards for studies in forensic genetics. *Forensic Science International*, *165*, 238–243.

Swartz, M. E., & Krull, I. S. (1997). *Analytical method development and validation*. New York: Marcel Dekker.
Taylor, J. K. (1981). Quality assurance of chemical measurements. *Analytical Chemistry*, *53*, 1588A–1596A.
Taylor, J. K. (1983). Validation of analytical methods. *Analytical Chemistry*, *55*, 600A–608A.
Taylor, J. K. (1987). *Quality assurance of chemical measurements*. Chelsea, Michigan: Lewis Publishers.
Taylor, J. K. (1990). *Statistical techniques for data analysis*. Chelsea, Michigan: Lewis Publishers.
VIM (2008). *International vocabulary of metrology—basic and general concepts and associated terms, VIM* (3rd ed.). JCGM 200:2008; also published as ISO Guide 99 (ISO/IEC Guide 99-12:2007). Available for download at <http://www.bipm.org/utils/common/documents/jcgm/JCGM_200_2008.pdf>.

Standards Organizations

American National Standards Institute (ANSI). <http://www.ansi.org/>.
American Society of Standards and Testing Materials (ASTM). <http://www.astm.org/>.
Bureau International des Poids et Mesures (BIPM). <http://www.bipm.org/>.
International Organization for Standardization <http://www.iso.org>.

Scientific Organizations

European Network of Forensic Science Institutes (ENFSI). <http://www.enfsi.eu>.
International Society of Forensic Genetics (ISFG). <http://www.isfg.org>.

ISFG Guidelines/Recommendations

Brinkmann, B., et al. (1989). Editorial: Recommendations of the Society for Forensic Haemogenetics Concerning DNA polymorphisms. *Forensic Science International*, *43*, 109–111.
Brinkmann, B., et al. (1992). Editorial: 1991 Report concerning recommendations of the DNA Commission of the International Society for Forensic Haemogenetics relating to the use of DNA polymorphisms. *Forensic Science International*, *52*, 125–130.
Bär, W., et al. (1992). Editorial: Recommendations of the DNA Commission of the International Society for Forensic Haemogenetics relating to the use of PCR-based polymorphisms. *Forensic Science International*, *55*, 1–3.
Bär, W., et al. (1993). Editorial: Statement by DNA Commission of the International Society for Forensic Haemogenetics concerning the National Academy of Sciences report on DNA technology in forensic science in the USA. *Forensic Science International*, *59*, 1–2.
Bär, W., et al. (1994). DNA recommendations—1994 report concerning further recommendations of the DNA Commission of the ISFH regarding PCR-based polymorphisms in STR (short tandem repeat) systems. *International Journal Legal Medicine*, *107*, 159–160.
Bär, W., et al. (1997). DNA recommendations—further report of the DNA Commission of the ISFH regarding the use of short tandem repeat systems. *International Journal Legal Medicine*, *110*, 175–176.
Carracedo, A., et al. (2000). DNA Commission of the International Society for Forensic Genetics: Guidelines for mitochondrial DNA typing. *Forensic Science International*, *110*, 79–85.
Gill, P., et al. (2001). DNA Commission of the International Society of Forensic Genetics: Recommendations on forensic analysis using Y-chromosome STRs. *Forensic Science International*, *124*, 5–10.
Gusmão, L., et al. (2006). DNA Commission of the International Society of Forensic Genetics (ISFG): An update of the recommendations on the use of Y-STRs in forensic analysis. *Forensic Science International*, *157*, 187–197.
Gill, P., et al. (2006). DNA Commission of the International Society of Forensic Genetics: Recommendations on the interpretation of mixtures. *Forensic Science International*, *160*, 90–101.
Gjertson, D. W., et al. (2007). ISFG: Recommendations on biostatistics in paternity testing. *Forensic Science International: Genetics*, *1*(3–4), 223–231.
Linacre, A., et al. (2011). ISFG: Recommendations regarding the use of non-human (animal) DNA in forensic genetic investigations. *Forensic Science International: Genetics* (in press, doi:10.1016/j.fsigen.2010.10.017).
Morling, N., et al. (2002). Paternity Testing Commission of the International Society of Forensic Genetics: Recommendations on genetic investigations in paternity cases. *Forensic Science International*, *129*, 148–157.
Prinz, M., et al. (2007). DNA Commission of the International Society for Forensic Genetics (ISFG): Recommendations regarding the role of forensic genetics for disaster victim identification (DVI). *Forensic Science International: Genetics*, *1*(1), 3–12.

SWGDAM/TWGDAM/DAB Guidelines

Budowle, B., et al. (1995). Guidelines for a quality assurance program for DNA analysis. *Crime Laboratory Digest*, *22*(2), 20–43.

DNA Advisory Board (2000). Statistical and population genetics issues affecting the evaluation of the frequency of occurrence of DNA profiles calculated from pertinent population database(s). *Forensic Science Communications*, *2*(3). (Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/dnastat.htm>.

Kearney, J., et al. (1989). Guidelines for a quality assurance program for DNA restriction fragment length polymorphism analysis. *Crime Laboratory Digest*, *16*(2), 40–59.

Kearney, J., et al. (1991). Guidelines for a quality assurance program for DNA analysis. *Crime Laboratory Digest*, *18*(2), 44–75.

SWGDAM (2000). Short tandem repeat (STR) interpretation guidelines. *Forensic Science Communications*, *2*(3) Available at http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/strig.htm>.

SWGDAM (2001). Training guidelines. *Forensic Science Communications*, *3*(4). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/oct2001/kzinski.htm>

SWGDAM (2003). Bylaws of the scientific working group on DNA analysis methods. *Forensic Science Communications*, *5*(2). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/april2003/swgdambylaws.htm>.

SWGDAM (2003). Guidelines for mitochondrial DNA (mtDNA) nucleotide sequence interpretation. *Forensic Science Communications*, *5*(2). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/april2003/swgdammitodna.htm>.

SWGDAM (2003). Guidance document for implementing health and safety programs in DNA laboratories. *Forensic Science Communications*, *5*(2). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/april2003/swgdam-safety.htm>.

SWGDAM (2004). Revised validation guidelines. *Forensic Science Communications*, *6*(3). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/july2004/standards/2004_03_standards02.htm>.

SWGDAM (2004). Report on the current activities of the scientific working group on DNA analysis methods Y-STR subcommittee. *Forensic Science Communications*, *6*(3). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/july2004/standards/2004_03_standards03.htm>.

SWGDAM (2009). Y-chromosome short tandem repeat (Y-STR) interpretation guidelines. *Forensic Science Communications*, *11*(1). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/jan2009/standards/2009_01_standards01.htm>.

SWGDAM (2009). SWGDAM recommendations to the FBI director on the "Interim plan for the release of information in the event of a 'partial match' at NDIS". *Forensic Science Communications*, *11*(4). Available at <http://www2.fbi.gov/hq/lab/fsc/backissu/oct2009/standard_guidlines/swgdam.html>.

SWGDAM. (2010). *SWGDAM interpretation guidelines for autosomal STR typing by forensic DNA testing laboratories*. Available at <http://www.fbi.gov/about-us/lab/codis/swgdam-interpretation-guidelines>.

Quality Assurance Standards

Federal Bureau of Investigation (2000). Quality assurance standards for forensic DNA testing laboratories. *Forensic Science Communications*, *2*(3). Available at: <http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/codispre.htm> or <http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/codis2a.htm>.

Federal Bureau of Investigation (2000). Quality assurance standards for convicted offender DNA databasing laboratories. *Forensic Science Communications*, *2*(3). Available at: <http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/codispre.htm> or <http://www2.fbi.gov/hq/lab/fsc/backissu/july2000/codis1a.htm>.

Presley, L. A. (1999). The evolution of quality assurance standards for forensic DNA analyses in the United States. *Profiles in DNA*, *3*(2), 10–11. Available at <http://www.promega.com/profiles>.

Quality Assurance Standards (QAS) for Forensic DNA Testing Laboratories. (2009). Available at <http://www2.fbi.gov/hq/lab/html/testinglab.htm> or <http://www.fbi.gov/about-us/lab/codis/qas_testlabs>.

Quality Assurance Standards (QAS) for DNA Databasing Laboratories. (2009). Available at <http://www.fbi.gov/about-us/lab/codis/qas_databaselabs>.

Laboratory Accreditation

American Society of Crime Laboratory Directors-Laboratory Accreditation Board (ASCLD-LAB). <http://www.ascld-lab.org/>.

Forensic Quality Services-International (FQS-I). <http://www.forquality.org>.
ISO/IEC 17025:2005 (2005) General requirements for the competence of testing and calibration laboratories. Available from <http://www.iso.org>.
Malkoc, E., & Neuteboom, W. (2007). The current status of forensic science laboratory accreditation in Europe. *Forensic Science International*, *167*, 121–126.

Proficiency Testing

Collaborative Testing Services (CTS). <http://www.cts-interlab.com/>.
College of American Pathologists (CAP). <http://www.cap.org>.
GEDNAP. <http://www.gednap.org/>.
National Research Council (1996). *The evaluation of forensic DNA evidence*. Washington, DC: National Academy Press.
Peterson, J. L., et al. (2003). The feasibility of external blind DNA proficiency testing: I. Background and findings. *Journal of Forensic Sciences*, *48*, 21–31.
Peterson, J. L., et al. (2003). The feasibility of external blind DNA proficiency testing: II. Experience with actual blind tests. *Journal of Forensic Sciences*, *48*, 32–40.
Rand, S., et al. (2002). The GEDNAP (German DNA profiling group) blind trial concept. *International Journal of Legal Medicine*, *116*, 199–206.
Rand, S., et al. (2004). The GEDNAP blind trial concept part II. Trends and developments. *International Journal of Legal Medicine*, *118*, 83–89.
Serological Research Institute. <http://www.serological.com/>.

Validation

Buckleton, J. (2009). Validation issues around DNA typing of low level DNA. *Forensic Science International: Genetics*, *3*, 255–260.
Budowle, B., et al. (2008). Criteria for validation of methods in microbial forensics. *Applied and Environmental Microbiology*, *74*, 5599–5607.
Butler, J. M., et al. (2004). Can the validation process in forensic DNA typing be standardized? In *Proceedings of the 15th international symposium on human identification*. Available at <http://www.promega.com/geneticidproc/ussymp15proc/oralpresentations/butler.pdf>.
Butler, J. M. (2006). Debunking some urban legends surrounding validation within the forensic DNA community. *Profiles in DNA*, *9*(2), 3–6. Available at <http://www.promega.com/profiles/>.
Butler, J. M. (2007). Validation: what is it, why does it matter, and how should it be done?. *Forensic News* (Applied Biosystems newsletter), January 2007 issue. Available at <http://marketing.appliedbiosystems.com/mk/get/FORENSICNEWS_PAST_ISSUES>; see also <http://www.cstl.nist.gov/biotech/strbase/pub_pres/Butler_ForensicNews_Jan2007.pdf>.
ENFSI (2006). *Validation and implementation of (New) methods*. Available at <www.enfsi.eu/get_doc.php?uid = 144>.
ENFSI DNA Working Group (2010). *Recommended minimum criteria for the validation of various aspects of the DNA profiling process*. Available at <http://www.enfsi.eu>.
EURACHEM Guide. (1998). *The fitness for purpose of analytical methods: a laboratory guide to method validation and related topics*. Available at <http://www.eurachem.org/guides/pdf/valid.pdf>.
Feinberg, M., et al. (2004). New advances in method validation and measurement uncertainty aimed at improving the quality of chemical data. *Analytical and Bioanalytical Chemistry*, *380*, 502–514.
Green, J. M. (1996). A practical guide to analytical method validation. *Analytical Chemistry*, *68*, 305A–309A.
Gunzler, H. (Ed.). (1996). *Accreditation and quality assurance in analytical chemistry*. New York: Springer.
Huber, L. (1999). *Validation and qualification in analytical laboratories*. Boca Raton: CRC Press.
Peters, F. T., et al. (2007). Validation of new methods. *Forensic Science International*, *165*, 216–224.
Promega Corporation. (2006). Internal validation of STR systems reference manual. Available at <http://www.promega.com/techserv/apps/hmnid/referenceinformation/powerplex/ValidationManual.pdf>.
Validation information on STRBase. <http://www.cstl.nist.gov/biotech/strbase/validation.htm>.
Vincent, E. B., & Bessetti, J. (2003). Validation questions and answers. *Profiles in DNA*, *6*(2), 13–14. Available at <http://www.promega.com/profiles>.

Measurement Uncertainty

Bell, S. (1999). A beginner's guide to uncertainty of measurement. *Measurement Good Practice Guide*, *11*(2). National Physical Laboratory: Teddington, UK. Available at <http://www.wmo.int/pages/prog/gcos/documents/gruanmanuals/UK_NPL/mgpg11.pdf>.

NIST Reference on constants, units, and uncertainty. <http://www.physics.nist.gov/cuu/Uncertainty/index.html>.

Salyards, J. (2008, Feb/Mar). Estimating the uncertainty. *Forensic Magazine*, 42–44. Available at <http://www.forensicmag.com/article/estimating-uncertainty>.

Taylor, B. N., & Kuyatt, C. E. (1994). Guidelines for evaluating and expressing the uncertainty of NIST measurement results. *NIST Technical Note 1297*. Available at <http://physics.nist.gov/Pubs/guidelines/TN1297/tn1297s.pdf>.

VIM (2008). *International vocabulary of metrology—basic and general concepts and associated terms*, *VIM* (3rd ed.), JCGM 200:2008; also published as ISO Guide 99 (ISO/IEC Guide 99-12:2007). Available for download at: <http://www.bipm.org/utils/common/documents/jcgm/JCGM_200_2008.pdf>.

Developmental Validation Studies

Barbisin, M., et al. (2009). Developmental validation of the Quantifiler Duo DNA Quantification kit for simultaneous quantification of total human and human male DNA and detection of PCR inhibitors in biological samples. *Journal of Forensic Sciences*, *54*, 305–319.

Brevnov, M. G., et al. (2009). Developmental validation of the PrepFiler Forensic DNA Extraction Kit for extraction of genomic DNA from biological samples. *Journal of Forensic Sciences*, *54*, 599–607.

Budowle, B., et al. (1997). Validation studies of the CTT STR multiplex system. *Journal of Forensic Sciences*, *42*(4), 701–707.

Buel, E., et al. (1995). PCR amplification of animal DNA with human X-Y amelogenin primers used in gender determination. *Journal of Forensic Sciences*, *40*, 641–644.

Caragine, T., et al. (2009). Validation of testing and interpretation protocols for low template DNA samples using AmpFlSTR Identifiler. *Croatian Medical Journal*, *50*, 250–267.

Collins, P. J., et al. (2004). Developmental validation of a single-tube amplification of the 13 CODIS STR loci, D2S1338, D19S433, and amelogenin: the AmpFlSTR Identifiler PCR amplification kit. *Journal of Forensic Sciences*, *49*, 1265–1277.

Coticone, S. R., et al. (2004). Development of the AmpFISTR SEfiler PCR amplification kit: A new multiplex containing the highly discriminating ACTBP2 (SE33) locus. *International Journal of Legal Medicine*, *118*, 224–234.

Cotton, E. A., et al. (2000). Validation of the AMPFlSTR SGM Plus system for use in forensic casework. *Forensic Science International*, *112*, 151–161.

Crouse, C., & Schumm, J. W. (1995). Investigation of species specificity using nine PCR-based human STR systems. *Journal of Forensic Sciences*, *40*(6), 952–956.

Daniels, D. L., et al. (2004). SWGDAM developmental validation of a 19-locus Y-STR system for forensic casework. *Journal of Forensic Sciences*, *49*, 668–683.

Dawnay, N., et al. (2008). A forensic STR profiling system for the Eurasian badger: A framework for developing profiling systems for wildlife species. *Forensic Science International: Genetics*, *2*, 47–53.

Dayton, M., et al. (2009). Developmental validation of short tandem repeat reagent kit for forensic DNA profiling of canine biological material. *Croatian Medical Journal*, *50*, 268–285.

Dixon, L. A., et al. (2005). Validation of a 21-locus autosomal SNP multiplex for forensic identification purposes. *Forensic Science International*, *154*, 62–77.

Ensenberger, M. G., et al. (2010). Developmental validation of the PowerPlex 16 HS System: An improved 16-locus fluorescent STR multiplex. *Forensic Science International: Genetics*, *4*, 257–264.

Fox, J. C., et al. (2003). Development, characterization, and validation of a sensitive primate-specific quantification assay for forensic analysis. *Biotechniques*, *34*, 314–322.

Frazier, R. R., et al. (1996). Validation of the Applied Biosystems Prism™ 377 automated sequencer for forensic short tandem repeat analysis. *Electrophoresis*, *17*, 1550–1552.

Frégeau, C. J., et al. (1999). Validation of highly polymorphic fluorescent multiplex short tandem repeat systems using two generations of DNA sequencers. *Journal of Forensic Sciences*, *44*, 133–166.

Frégeau, C. J., et al. (2003). AmpFlSTR Profiler Plus short tandem repeat DNA analysis of casework samples, mixture samples, and nonhuman DNA samples amplified under reduced PCR volume conditions (25 microL). *Journal of Forensic Sciences*, *48*(5), 1014–1034.

Gill, P., et al. (2001). Sizing short tandem repeat alleles in capillary array gel electrophoresis instruments. *Electrophoresis*, *22*, 2670–2678.

Green, R. L., et al. (2005). Developmental validation of the Quantifiler real-time PCR kits for the quantification of human nuclear DNA samples. *Journal of Forensic Sciences*, *50*, 809–825.

Greenspoon, S. A., et al. (2004). Application of the BioMek 2000 Laboratory Automation Workstation and the DNA IQ System to the extraction of forensic casework samples. *Journal of Forensic Sciences*, *49*, 29–39.

Greenspoon, S. A., et al. (2004). Validation and implementation of the PowerPlex 16 BIO System STR multiplex for forensic casework. *Journal of Forensic Sciences*, *49*, 71–80.

Holland, M. M., & Parsons, T. J. (1999). Mitochondrial DNA sequence analysis – validation and use for forensic casework. *Forensic Science Review*, *11*(1), 22–50.

Holt, C. L., et al. (2002). TWGDAM validation of AmpFlSTR PCR amplification kits for forensic DNA casework. *Journal of Forensic Sciences*, *47*, 66–96.

Howard, C., et al. (2008). Developmental validation of a Cannabis sativa STR multiplex system for forensic analysis. *Journal of Forensic Sciences*, *53*, 1061–1067.

Hudlow, W. R., et al. (2008). A quadruplex real-time qPCR assay for the simultaneous assessment of total human DNA, human male DNA, DNA degradation and the presence of PCR inhibitors in forensic samples: a diagnostic tool for STR typing. *Forensic Science International: Genetics*, *2*, 108–125.

Johnson, C. L., et al. (2003). Validation and uses of a Y-chromosome STR 10-plex for forensic and paternity laboratories. *Journal of Forensic Sciences*, *48*(6), 1260–1268.

Junge, A., & Madea, B. (1999). Validation studies and characterization of variant alleles at the short tandem repeat locus D12S391. *International Journal of Legal Medicine*, *112*, 67–69.

Junge, A., et al. (2003). Validation of the multiplex kit genRESMPX-2 for forensic casework analysis. *International Journal of Legal Medicine*, *117*, 317–325.

Kadash, K., et al. (2004). Validation study of the TrueAllele automated data review system. *Journal of Forensic Sciences*, *49*, 660–667.

Kimpton, C. P., et al. (1994). Evaluation of an automated DNA profiling system employing multiplex amplification of four tetrameric STR loci. *International Journal of Legal Medicine*, *106*, 302–311.

Kimpton, C. P., et al. (1996). Validation of highly discriminating multiplex short tandem repeat amplification systems for individual identification. *Electrophoresis*, *17*(8), 1283–1293.

Koumi, P., et al. (2004). Evaluation and validation of the ABI 3700, ABI 3100, and the MegaBACE 1000 capillary array electrophoresis instruments for use with short tandem repeat microsatellite typing in a forensic environment. *Electrophoresis*, *25*, 2227–2241.

Krenke, B. E., et al. (2002). Validation of a 16-locus fluorescent multiplex system. *Journal of Forensic Sciences*, *47*, 773–785.

Krenke, B. E., et al. (2005). Validation of a male-specific, 12-locus fluorescent short tandem repeat (STR) multiplex. *Forensic Science International*, *148*, 1–14.

Krenke, B. E., et al. (2008). Developmental validation of a real-time PCR assay for the simultaneous quantification of total human and male DNA. *Forensic Science International Genetics*, *3*, 14–21.

Lazaruk, K., et al. (1998). Genotyping of forensic short tandem repeat (STR) systems based on sizing precision in a capillary electrophoresis instrument. *Electrophoresis*, *19*, 86–93.

Levedakou, E. N., et al. (2002). Characterization and validation studies of PowerPlex 2.1, a nine-locus short tandem repeat (STR) multiplex system and Penta D monoplex. *Journal of Forensic Sciences*, *47*, 757–772.

Leibelt, C., et al. (2003). Identification of a D8S1179 primer binding site mutation and the validation of a primer designed to recover null alleles. *Forensic Science International*, *133*, 220–227.

Lindquist, C. D., et al. (2011). Developmental validation of feline, bovine, equine, and cervid quantitative PCR assays. *Journal of Forensic Sciences*, *56*(Suppl.), S29–S35.

Mandrekar, M. N., et al. (2001). Development of a human DNA quantitation system. *Croatian Medical Journal*, *42*(3), 336–339.

Micka, K. A., et al. (1996). Validation of multiplex polymorphic STR amplification sets developed for personal identification applications. *Journal of Forensic Sciences*, *41*(4), 582–590.

Micka, K. A., et al. (1999). TWGDAM validation of a nine-locus and a four-locus fluorescent STR multiplex system. *Journal of Forensic Sciences*, *44*, 1243–1257.

Morley, J. M., et al. (1999). Validation of mitochondrial DNA mini-sequencing for forensic casework. *International Journal of Legal Medicine*, *112*, 241–248.

Mulero, J. J., et al. (2006). Development and validation of the AmpFlSTR Yfiler PCR Amplification kit: A male specific, single amplification 17 Y-STR multiplex system. *Journal of Forensic Sciences*, *51*, 64–75.

Mulero, J. J., et al. (2008). Development and validation of the AmpFlSTR MiniFiler PCR Amplification kit: A mini-STR multiplex for the analysis of degraded and/or PCR inhibited DNA. *Journal of Forensic Sciences*, *53*, 838–852.

Muller, K., et al. (2009). Validation of the short amplicon multiplex Q8 including the German DNA database systems. *Journal of Forensic Sciences*, *54*, 862–865.

Musgrave-Brown, E., et al. (2007). Forensic validation of the SNPforID 52-plex assay. *Forensic Science International: Genetics*, *1*, 186–190.

Nicklas, J. A., & Buel, E. (2003). Development of an Alu-based, real-time PCR method for quantitation of human DNA in forensic samples. *Journal of Forensic Sciences*, *48*(5), 936–944.

Old, J. B., et al. (2009). Developmental validation of RSID-saliva: A lateral flow immunochromatographic strip test for the forensic detection of saliva. *Journal of Forensic Sciences*, *54*, 866–873.

Opel, K. L., et al. (2007). Developmental validation of reduced-size STR miniplex primer sets. *Journal of Forensic Sciences*, *52*, 1263–1271.

Prinz, M., et al. (2001). Validation and casework application of a Y chromosome specific STR multiplex. *Forensic Science International*, *120*, 177–188.

Richard, M. L., et al. (2003). Developmental validation of a real-time quantitative PCR assay for automated quantification of human DNA. *Journal of Forensic Sciences*, *48*(5), 1041–1046.

Schweers, B. A., et al. (2008). Developmental validation of a novel lateral flow strip test for rapid identification of human blood (Rapid Stain Identification—Blood). *Forensic Science International: Genetics*, *2*, 243–247.

Shewale, J. G., et al. (2004). Y-chromosome STR system, Y-PLEX 12, for forensic casework: Development and validation. *Journal of Forensic Sciences*, *49*(6), 1278–1290.

Sinha, S. K., et al. (2003). Development and validation of a multiplexed Y-chromosome STR genotyping system, Y-PLEX 6, for forensic casework. *Journal of Forensic Sciences*, *48*(1), 93–103.

Sinha, S. K., et al. (2003). Development and validation of the Y-PLEX 5, a Y-chromosome STR genotyping system, for forensic casework. *Journal of Forensic Sciences*, *48*(5), 985–1000.

Spathis, R., & Lum, J. K. (2008). An updated validation of Promega's PowerPlex 16 System: High throughput databasing under reduced PCR volume conditions on Applied Biosystem's 96 capillary 3730xl DNA Analyzer. *Journal of Forensic Sciences*, *53*, 1353–1357.

Tucker, V. C., et al. (2011). Developmental validation of the PowerPlex ESI 16 and PowerPlex ESI 17 Systems: STR multiplexes for the new European standard. *Forensic Science International: Genetics* (in press) (doi:10.1016/j.fsigen.2010.03.014).

Turrina, S., et al. (2007). Development and forensic validation of a new multiplex PCR assay with 12 X-chromosomal short tandem repeats. *Forensic Science International: Genetics*, *1*, 201–204.

Wallin, J. M., et al. (1998). TWGDAM validation of the AmpFISTR Blue PCR amplification kit for forensic casework analysis. *Journal of Forensic Sciences*, *43*, 854–870.

Wallin, J. M., et al. (2002). Constructing universal multiplex PCR systems for comparative genotyping. *Journal of Forensic Sciences*, *47*(1), 52–65.

Walsh, S., et al. (2011). Developmental validation of the IrisPlex system: Determination of blue and brown iris colour for forensic intelligence. *Forensic Science International: Genetics* (in press) (doi:10.1016/j.fsigen.2010.09.008).

Wilson, M. R., et al. (1995). Validation of mitochondrial DNA sequencing for forensic casework analysis. *International Journal of Legal Medicine*, *108*, 68–74.

Internal Validation Studies

Andersen, J. F., et al. (1996). Further validation of a multiplex STR system for use in routine forensic identity testing. *Forensic Science International*, *78*, 47–64.

Buse, E. L., et al. (2003). Performance evaluation of two multiplexes used in fluorescent short tandem repeat DNA analysis. *Journal of Forensic Sciences*, *48*(2), 348–357.

Clayton, T. M., et al. (1995). Further validation of a quadruplex STR DNA typing system: a collaborative effort to identify victims of a mass disaster. *Forensic Science International*, *76*, 17–25.

Crouse, C. A. (2001). Implementation of forensic DNA analysis on casework evidence at the Palm Beach County Sheriff's Office Crime Laboratory: Historical perspective. *Croatian Medical Journal*, *42*(3), 247–251.

Drábek, J. (2009). Validation of software for calculating the likelihood ratio for parentage and kinship. *Forensic Science International: Genetics*, *3*, 112–118.

Frank, W. E., et al. (2001). Validation of the AmpFlSTR Profiler Plus PCR amplification kit for use in forensic casework. *Journal of Forensic Sciences*, *46*(3), 642–646.
Gaines, M. L., et al. (2002). Reduced volume PCR amplification reactions using the AmpFlSTR Profiler Plus kit. *Journal of Forensic Sciences*, *47*(6), 1224–1237.
Greenspoon, S. A., et al. (2000). Validation of the PowerPlex 1.1 loci for use in human identification. *Journal of Forensic Sciences*, *45*(3), 677–683.
Gross, A. M., et al. (2008). Internal validation of the AmpFlSTR Yfiler amplification kit for use in forensic casework. *Journal of Forensic Sciences*, *53*, 125–134.
Isenberg, A. R., et al. (1998). Analysis of two multiplexed short tandem repeat systems using capillary electrophoresis with multiwavelength florescence detection. *Electrophoresis*, *19*, 94–100.
Jarman, P. G., et al. (2009). Mitochondrial DNA validation in a state laboratory. *Journal of Forensic Sciences*, *54*, 95–102.
LaFountain, M., et al. (1998). Validation of capillary electrophoresis for analysis of the X-Y homologous amelogenin gene. *Journal of Forensic Sciences*, *43*(6), 1188–1194.
LaFountain, M. J., et al. (2001). TWGDAM validation of the AmpFlSTR Profiler Plus and AmpFlSTR COfiler STR multiplex systems using capillary electrophoresis. *Journal of Forensic Sciences*, *46*(5), 1191–1198.
Luce, C., et al. (2009). Validation of the AMPFlSTR MiniFiler PCR amplification kit for use in forensic casework. *Journal of Forensic Sciences*, *54*, 1046–1054.
Lygo, J. E., et al. (1994). The validation of short tandem repeat (STR) loci for use in forensic casework. *International Journal of Legal Medicine*, *107*, 77–89.
Moretti, T. R., et al. (2001). Validation of short tandem repeats (STRs) for forensic usage: performance testing of fluorescent multiplex STR systems and analysis of authentic and simulated forensic samples. *Journal of Forensic Sciences*, *46*(3), 647–660.
Moretti, T. R., et al. (2001). Validation of STR typing by capillary electrophoresis. *Journal of Forensic Sciences*, *46*(3), 661–676.
Nelson, M. S., et al. (2002). Detection of a primer-binding site polymorphism for the STR locus D16S539 using the Powerplex 1.1 system and validation of a degenerate primer to correct for the polymorphism. *Journal of Forensic Sciences*, *47*(2), 345–349.
Pawlowski, R., & Maciejewska, A. (2000). Forensic validation of a multiplex containing nine STRs—population genetics in northern Poland. *International Journal of Legal Medicine*, *114*, 45–49.
Pestoni, C., et al. (1995). The use of the STRs HUMTH01, HUMVWA31/A, HUMF13A1, HUMFES/FPS, HUMLPL in forensic application: Validation studies and population data for Galicia (NW Spain). *International Journal of Legal Medicine*, *107*, 283–290.
Potter, T. (2003). Co-amplification of ENFSI-loci D3S1358, D8S1179 and D18S51: Validation of new primer sequences and allelic distribution among 2874 individuals. *Forensic Science International*, *138*, 104–110.
Ryan, J. H., et al. (2004). The application of an automated allele concordance analysis system (CompareCalls) to ensure the accuracy of single-source STR DNA profiles. *Journal of Forensic Sciences*, *49*(3), 492–499.
Sgueglia, J. B., et al. (2003). Precision studies using the ABI Prism 3100 genetic analyzer for forensic DNA analysis. *Analytical and Bioanalytical Chemistry*, *376*, 1247–1254.
Sparkes, R., et al. (1996). The validation of a 7-locus multiplex STR test for use in forensic casework. (I). Mixtures, ageing, degradation and species studies. *International Journal of Legal Medicine*, *109*, 186–194.
Sparkes, R., et al. (1996). The validation of a 7-locus multiplex STR test for use in forensic casework. (II), Artefacts, casework studies and success rates. *International Journal of Legal Medicine*, *109*, 195–204.
Tomsey, C. S., et al. (2001). Comparison of PowerPlex 16, PowerPlex 1.1/2.1, and ABI AmpFlSTR Profiler Plus/COfiler for forensic use. *Croatian Medical Journal*, *42*(3), 239–243.
van Oorschot, R. A., et al. (1996). HUMTH01 validation studies: effect of substrate, environment, and mixtures. *Journal of Forensic Sciences*, *41*(1), 142–145.
Westring, C. G., et al. (2007). Validation of reduced-scale reactions for the Quantifiler Human DNA kit. *Journal of Forensic Sciences*, *52*, 1035–1043.
Wiegand, P., et al. (1993). Forensic validation of the STR systems SE 33 and TC 11. *International Journal of Legal Medicine*, *105*, 315–320.

Interlaboratory Studies

Summary of interlab studies. <http://www.cstl.nist.gov/biotech/strbase/interlab.htm>.

EDNAP (20)

Andersen, J. F., et al. (1996). Report on the third EDNAP collaborative STR exercise. *Forensic Science International*, *78*, 83–93.

Bjerre, A., et al. (1997). A report of the 1995 and 1996 paternity testing workshops of the english speaking working group of the International Society of Forensic Haemogenetics. *Forensic Science International*, *90*, 41–55.

Brion, M., et al. (2005). A collaborative study of the EDNAP group regarding Y-chromosome binary polymorphism analysis. *Forensic Science International*, *153*, 103–108.

Carracedo, A., et al. (1998). Reproducibility of mtDNA analysis between laboratories: A report of the European DNA Profiling group (EDNAP). *Forensic Science International*, *97*, 165–170.

Carracedo, A., et al. (2001). Results of a collaborative study of the EDNAP group regarding the reproducibility and robustness of the Y-chromosome STRs DYS19, DYS389 I and II, DYS390 and DYS393 in a PCR pentaplex format. *Forensic Science International*, *119*, 28–41.

Dixon, L. A., et al. (2006). Analysis of artificially degraded DNA using STRs and SNPs–results of a collaborative European (EDNAP) exercise. *Forensic Science International*, *164*, 33–44.

Gill, P., et al. (1992). A report of an international collaborative experiment to demonstrate the uniformity obtainable using DNA profiling techniques. *Forensic Science International*, *53*, 29–43.

Gill, P., et al. (1994). Report of the European DNA profiling group (EDNAP) – towards standardization of short tandem repeat (STR) loci. *Forensic Science International*, *65*, 51–59.

Gill, P., et al. (1997). Report of the European DNA profiling group (EDNAP): An investigation of the complex STR loci D21S11 and HUMFIBRA (FGA). *Forensic Science International*, *86*, 25–33.

Gill, P., et al. (1998). Report of the European DNA Profiling group (EDNAP) – an investigation of the hypervariable loci ACTBP2, APOA11 and D11S554 and the compound loci D12S391 and D1S1656. *Forensic Science International*, *98*, 193–200.

Haas, C., et al. (2011). mRNA profiling for the identification of blood-Results of a collaborative EDNAP exercise. *Forensic Science International: Genetics*, *5*, 21–26.

Kimpton, C. P., et al. (1995). Report on the second EDNAP collaborative STR exercise. *Forensic Science International*, *71*, 137–152.

Parson, W., et al. (2004). The EDNAP mitochondrial DNA population database (EMPOP) collaborative exercises: Organisation, results and perspectives. *Forensic Science International*, *139*, 215–226.

Parson, W., et al. (2008). Identification of West Eurasian mitochondrial haplogroups by mtDNA SNP screening: Results of the 2006-2007 EDNAP collaborative exercise. *Forensic Science International: Genetics*, *2*, 61–68.

Sanchez, J. J., et al. (2008). Forensic typing of autosomal SNPs with a 29 SNP-multiplex–results of a collaborative EDNAP exercise. *Forensic Science International: Genetics*, *2*, 176–183.

Schneider, P. M., et al. (1991). Report of a European collaborative exercise comparing DNA typing results using a single locus VNTR probe. *Forensic Science International*, *49*, 1–15.

Schneider, P. M., et al. (1999). Results of a collaborative study regarding the standardization of the Y-linked STR system DYS385 by the European DNA Profiling (EDNAP) group. *Forensic Science International*, *102*, 159–165.

Schneider, P. M., et al. (2004). STR analysis of artificially degraded DNA—results of a collaborative European exercise. *Forensic Science International*, *139*, 123–134.

Tomas, C., et al. (2011). Autosomal SNP typing of forensic samples with the GenPlex HID System: Results of a collaborative study. *Forensic Science International: Genetics* (in press) (doi:10.1016/j.fsigen.2010.06.007).

Tully, G., et al. (2004). Results of a collaborative study of the EDNAP group regarding mitochondrial DNA heteroplasmy and segregation in hair shafts. *Forensic Science International*, *140*, 1–11.

Welch, L., et al. (2011). A comparison of mini-STRs versus standard STRs-Results of a collaborative European (EDNAP) exercise. *Forensic Science International: Genetics*, *5*, 257–258.

English Speaking Working Group-ISFG (5)

Bjerre, A., et al. (1997). A report of the 1995 and 1996 paternity testing workshops of the English Speaking Working Group of the International Society for Forensic Haemogenetics. *Forensic Science International*, *90*, 41–55.

Hallenberg, C., & Morling, N. (2001). A report of the 1997, 1998 and 1999 paternity testing workshops of the English Speaking Working Group of the International Society for Forensic Genetics. *Forensic Science International*, *116*, 23–33.

Hallenberg, C., & Morling, N. (2002). A report of the 2000 and 2001 paternity testing workshops of the English Speaking Working Group of the International Society for Forensic Genetics. *Forensic Science International*, *129*, 43–50.

Syndercombe-Court, D., & Lincoln, P. (1995). A review of the 1991-1994 paternity testing workshops of the English-Speaking Working Group. In A. Carracedo, B. Brinkmann & W. Bär (Eds.), *Advances in forensic haemogenetics 6* (pp. 683–685). Berlin: Springer-Verlag.

Thomsen, A. R., et al. (2009). A report of the 2002-2008 paternity testing workshops of the English Speaking Working Group of the International Society for Forensic Genetics. *Forensic Science International: Genetics*, *3*, 214–221.

GEP-ISFG (15)

Alonso, A., et al. (2002). Results of the 1999-2000 collaborative exercise and proficiency testing program on mitochondrial DNA of the GEP-ISFG: An inter-laboratory study of the observed variability in the heteroplasmy level of hair from the same donor. *Forensic Science International*, *125*, 1–7.

Crespillo, M., et al. (2006). Results of the 2003-2004 GEP-ISFG collaborative study on mitochondrial DNA: Focus on the mtDNA profile of a mixed semen-saliva stain. *Forensic Science International*, *160*, 157–167.

Gómez, J., et al. (1997). A review of the collaborative exercises on DNA typing of the Spanish and Portuguese ISFH Working Group. International Society for Forensic Haemogenetics. *International Journal of Legal Medicine*, *110*, 273–277.

Gómez, J., & Carracedo, A. (2000). The 1998-1999 collaborative exercises and proficiency testing program on DNA typing of the Spanish and Portuguese Working Group of the International Society for Forensic Genetics (GEP-ISFG). *Forensic Science International*, *114*, 21–30.

Gusmão, L., et al. (2003). Results of the GEP-ISFG collaborative study on the Y chromosome STRs GATA A10, GATA C4, GATA H4, DYS437, DYS438, DYS439, DYS460 and DYS461: Population data. *Forensic Science International*, *135*, 150–157.

Gusmão, L., et al. (2005). Mutation rates at Y chromosome specific microsatellites. *Human Mutation*, *26*, 520–528.

Gusmão, L., et al. (2009). A GEP-ISFG collaborative study on the optimization of an X-STR decaplex: Data on 15 Iberian and Latin American populations. *International Journal of Legal Medicine*, *123*, 227–234.

Montesino, M., et al. (2007). Analysis of body fluid mixtures by mtDNA sequencing: An inter-laboratory study of the GEP-ISFG working group. *Forensic Science International*, *168*, 42–56.

Prieto, L., et al. (2003). The 2000-2001 GEP-ISFG Collaborative Exercise on mtDNA: Assessing the cause of unsuccessful mtDNA PCR amplification of hair shaft samples. *Forensic Science International*, *134*, 46–53.

Prieto, L., et al. (2008). 2006 GEP-ISFG collaborative exercise on mtDNA: Reflections about interpretation, artifacts, and DNA mixtures. *Forensic Science International: Genetics*, *2*, 126–133.

Prieto, L., et al. (2010). The GHEP-EMPOP collaboration on mtDNA population data-A new resource for forensic casework. *Forensic Science International Genetics* (in press)

Salas, A., et al. (2005). Mitochondrial DNA error prophylaxis: Assessing the causes of errors in the GEP'02-03 proficiency testing trial. *Forensic Science International*, *148*, 191–198.

Sánchez-Diz, P., et al. (2003). Results of the GEP-ISFG collaborative study on two Y-STRs tetraplexes: GEPY I (DYS461, GATA C4, DYS437 and DYS438) and GEPY II (DYS460, GATA A10, GATA H4 and DYS439). *Forensic Science International*, *135*, 158–162.

Sánchez-Diz, P., et al. (2008). Population and segregation data on 17 Y-STRs: Results of a GEP-ISFG collaborative study. *International Journal of Legal Medicine*, *122*, 529–533.

van Asch, B., et al. (2009). Forensic analysis of dog (Canis lupus familiaris) mitochondrial DNA sequences: an inter-laboratory study of the GEP-ISFG working group. *Forensic Science International: Genetics*, *4*, 49–54.

NIST (13)

Butler, J. M., & Kline, M. C. (2005). NIST Mixture interpretation interlaboratory study 2005 (MIX05). In *Proceedings of the 16th international symposium on human identification*. Available at <http://www.cstl.nist.gov/biotech/strbase/interlab/MIX05/MIX05poster.pdf>.

Duewer, D. L., et al. (1995). Interlaboratory comparison of autoradiographic DNA profiling measurements. 2. Measurement uncertainty and its propagation. *Analytical Chemistry*, *67*, 1220–1231.

Duewer, D. L., et al. (1997). Interlaboratory comparison of autoradiographic DNA profiling measurements. 4. Protocol effects. *Analytical Chemistry*, *69*, 1882–1892.

Duewer, D. L., et al. (1998). Interlaboratory comparison of autoradiographic DNA profiling measurements: Precision and concordance. *Journal of Forensic Sciences*, *43*, 465–471.

Duewer, D. L., et al. (2000). RFLP band size standards: NIST Standard Reference Material® 2390. *Journal of Forensic Sciences*, *45*, 1093–1105.
Duewer, D. L., et al. (2000). RFLP band size standards: Cell line K562 values from 1991—1997 proficiency studies. *Journal of Forensic Sciences*, *45*, 1106–1118.
Duewer, D. L., et al. (2001). NIST Mixed Stain Studies #1 and #2: interlaboratory comparison of DNA quantification practice and short tandem repeat multiplex performance with multiple-source samples. *Journal of Forensic Sciences*, *46*, 1199–1210.
Duewer, D. L., et al. (2004). NIST Mixed Stain Study #3: Signal intensity balance in commercial short tandem repeat multiplexes. *Analytical Chemistry*, *76*, 6928–6934.
Kline, M. C., et al. (1997). Interlaboratory evaluation of short tandem repeat triplex CTT. *Journal of Forensic Sciences*, *42*(5), 897–906.
Kline, M. C., et al. (2003). NIST mixed stain study 3: DNA quantitation accuracy and its influence on short tandem repeat multiplex signal intensity. *Analytical Chemistry*, *75*, 2463–2469.
Kline, M. C., et al. (2005). Results from the NIST 2004 DNA quantitation study. *Journal of Forensic Sciences*, *50*(3), 571–578.
Mudd, J. L., et al. (1994). Interlaboratory comparison of autoradiographic DNA profiling measurements. 1. Data and summary statistics. *Analytical Chemistry*, *66*, 3303–3317.
Stolorow, A. M., et al. (1996). Interlaboratory comparison of autoradiographic DNA profiling measurements. 3. Repeatability and reproducibility of RFLP band sizing, particularly bands of molecular size > 10 k base pairs. *Analytical Chemistry*, *68*, 1941–1947.

Reference Materials

Institute for Reference Materials and Measurements (IRMM). <http://irmm.jrc.ec.europa.eu/html/homepage.htm>.
National Institute of Standards and Technology (NIST). <http://www.nist.gov>.
Fregeau, C. J., et al. (1995). Characterization of human lymphoid cell lines GM9947 and GM9948 as intra- and interlaboratory reference standards for DNA typing. *Genomics*, *28*, 184–197.
Kline, M. C. (2006). NIST SRM Updates: Value-added to the current materials in SRM 2391b and SRM 2395. In *Poster at 17th international symposium on human identification*. Nashville, TN. (Available at <http://www.cstl.nist.gov/biotech/strbase/pub_pres/Promega2006_Kline.pdf>).
Kline, M. C., et al. (2009). Production and certification of NIST standard reference material 2372 human DNA quantitation standard. *Analytical & Bioanalytical Chemistry*, *394*, 1183–1192.
May, W. E., et al. (1999). Definitions of terms and modes used at NIST for value-assignment of reference materials for chemical measurements: *NIST Special Publication 260-136*. Washington DC: US Government Printing Office. Available at <http://www.cstl.nist.gov/div839/special_pubs/SP260136.pdf>.
NIST SRMs. <http://www.nist.gov/srm>
NIST SRM 2372 STRBase page. <http://www.cstl.nist.gov/strbase/srm2372.htm>.
Szibor, R., et al. (2003). Cell line DNA typing in forensic genetics—the necessity of reliable standards. *Forensic Science International*, *138*, 37–43.
Vallone, P. M., et al. (2008). Development and usage of a NIST standard reference material for real-time PCR quantitation of human DNA. *Forensic Science International Genetics Supplement Series*, *1*, 80–82.

STR Typing Kits and Concordance Studies

Budowle, B., & Sprecher, C. J. (2001). Concordance study on population database samples using the PowerPlex 16 kit and AmpFlSTR Profiler Plus kit and AmpFlSTR COfiler kit. *Journal of Forensic Sciences*, *46*, 637–641.
Budowle, B., et al. (2001). STR primer concordance study. *Forensic Science International*, *124*, 47–54.
Clayton, T. M., et al. (2004). Primer binding site mutations affecting the typing of STR loci contained within the AMPFlSTR SGM Plus kit. *Forensic Science International*, *139*, 255–259.
Hill, C. R., et al. (2010). Strategies for concordance testing. *Profiles in DNA (Promega)*, *13*(1). Available at <http://www.promega.com/profiles/>.
Kline, M. C., et al. (1998). Non-amplification of a vWA allele. *Journal of Forensic Sciences*, *43*(1), 250.
STRBase Null Allele Report. <http://www.cstl.nist.gov/biotech/strbase/NullAlleles.htm>.